Building and Environment 292 (2026) 114234

Contents lists available at ScienceDirect

Building and Environment

journal homepage: www.elsevier.com/locate/buildenv

Soundscapes in home environments: The impact of home types and soundscapes on the recovery benefits of psychophysiological stress

Chengmin Zhou [a,b,*], Mizhi Feng [a,b], Xuechen Zhang [a,b], Jake Kaner [c]

[a] College of Furnishings and Industrial Design, Nanjing Forestry University, Nanjing, 210037, Jiangsu, China
[b] Jiangsu Co-Innovation Center of Efficient Processing and Utilization of Forest Resources, Nanjing Forestry University, Nanjing, 210037, Jiangsu, China
[c] School of Art and Design, Nottingham Trent University, Nottingham, NG1 4FQ, United Kingdom

HIGHLIGHTS

- Natural and music sounds yield superior physiological and psychological recovery compared to urban soundscapes.
- Home environment moderates soundscape effects: the relaxed-oriented setting buffers the negative impact of urban sound, while the aesthetic-oriented setting supports positive arousal under natural sound.
- Revealing the audio-visual synergy recovery mechanism: Relaxed-oriented homes mitigate the negative effects of urban sound, while aesthetic-oriented homes enhance the positive awakening of natural sound.
- Notable gender differences: females show greater sensitivity to the recovery effects of natural and music sounds, while males' recovery effects depend on the type of home environment.

ARTICLE INFO

Keywords:
Home environment
Indoor soundscape
Audio-visual interaction
Stress recovery
Gender differences

ABSTRACT

Rapid urbanization has placed individuals under sustained high-stress loads, underscoring the urgent need for effective restorative environmental interventions. Using a 3×3 factorial experimental design, this study systematically examines how three types of soundscapes (natural, music, urban) and three types of home environments (functional-efficiency, aesthetic-display, and comfort-relaxation) influence stress recovery, and further explores the moderating role of gender. We assessed recovery using physiological indicators, including electroencephalography (EEG), electrocardiogram (ECG), and electrodermal activity (EDA), along with subjective rating scales, to evaluate the joint effects of sound–home interactions on both physiological and psychological restoration. The results show: (1) both natural soundscapes and music soundscapes significantly increased α, γ, and θ wave power, reduced β wave power, improved heart rate variability (HRV) indices, and enhanced perceived restoration, whereas urban soundscapes produced predominantly adverse effects; (2) home type modulated recovery, with the relaxation-oriented home most effectively buffering the adverse impact of urban soundscapes, the aesthetic-oriented home supporting stronger positive arousal under natural soundscapes, and the functional-oriented home showing the weakest restorative profile overall; (3) gender differences emerged in several physiological measures, with female participants responding more favorably to natural and music soundscapes, whereas male participants displayed a more context-dependent dual pattern of "recovery vs. arousal" across different home environments. These findings reveal the multidimensional mechanisms by which soundscapes and home environments jointly shape stress recovery and highlight the applied value of beneficial soundscapes (natural and musical) for indoor restorative design in everyday living spaces.

1. Introduction

In the context of rapid urbanization, individuals experience multiple sources of psychosocial stress arising from work demands, domestic pressures, and shifts in social structure [1]. Persistent high-stress exposure can trigger chronic stress reactions, manifesting as insomnia, headaches, and mood disorders, and in severe cases, contributing to cardiovascular and cerebrovascular risks [2]. Recent evidence shows

* Corresponding author at: College of Furnishings and Industrial Design, Nanjing Forestry University, Nanjing, 210037, Jiangsu, China.
Email addresses: zcm78@163.com (C. Zhou), fmz010914@163.com (M. Feng).

https://doi.org/10.1016/j.buildenv.2026.114234
Received 13 November 2025; Received in revised form 20 December 2025; Accepted 9 January 2026
Available online 16 January 2026
0360-1323/

that environmental factors—particularly acoustic load such as noise exposure—are critical determinants of urban health [3,4]. Traditional urban design strategies are increasingly insufficient to buffer these stressors, while environmental interventions, such as creating restorative spaces, have emerged as a promising approach to enhance psychological resilience and recovery capacity [5,6].

In recent years, the concept of "soundscapes" has attracted widespread attention across interdisciplinary fields. ISO 12,913–1 (2014) defines soundscapes as the perception or experience of the sound environment by people in specific situations, emphasizing the core role of subjective perception in evaluating sound environments rather than relying solely on physical noise indicators [7,8]. Traditional environmental acoustics research mainly focused on passive noise control—reducing harmful sound exposure through urban zoning and building sound insulation [9–11]. Although it achieved specific results, it ignored the potential role of positive sound experiences. In contrast, modern soundscapes research has shifted towards active and health-promoting orientations, not only avoiding adverse sounds but also emphasizing the creation of pleasant auditory environments [12,13], reflecting a paradigm shift from "noise avoidance" to "soundscapes enhancement" that aligns with public health trends. Empirical studies have shown that natural sounds and biophilic sound sources have positive effects, reducing stress and promoting recovery [14–17]; music can also lower physiological stress markers [18]. In urban environments, the reasonable use of music can create a pleasant atmosphere [19]. People have less preference for man-made sound sources, and natural soundscapes have a more positive effect on physiological responses [20]. This shift in research paradigm is of great significance for soundscapes intervention design, and in indoor environments such as homes, active soundscapes intervention is a feasible path to alleviate daily stress.

A soundscape emerges from the dynamic interaction among sound, environment, and listener, integrating physical acoustics, spatial context, and socio-cultural meaning. Consequently, soundscape perception is inherently multisensory rather than purely auditory. Prior work shows that coupling auditory and visual cues can enhance environmental appraisal, with auditory perception benefiting from visual input and vice versa [21,22]. Touhami et al. [23] reported that in university outdoor spaces, the acoustic environment strongly shaped perceived comfort, even more than thermal or lighting conditions, underscoring the importance of multisensory assessment in environmental design. Similarly, Van Renterghem and Lippens [24] found that adding natural sounds can relax the perception of an otherwise urban visual scene; even in the absence of visible nature, natural acoustic elements improved the perceived urban soundscape. The characteristics of the indoor environment at home are the interaction of auditory and visual stimuli, which suggests that in the indoor environment, sound scene intervention should be designed in coordination with visual features such as spatial materials, lighting, and color. Under this multi-sensory design perspective, the visual creation of restorative spaces mainly has three key elements: the biophilic nature of materials, the regulation of lighting and color, and the support of furniture systems for a sense of spatial control. This tripartite framework builds on environmental psychology and healing environment theories, which emphasize nature-based positive distractions, ambient conditions, and environmental control as core pathways through which interior design promotes stress coping and wellness [25–27]. Zhao et al. [28] showed that glass surfaces were perceived as more restorative than wood, whereas metal surfaces were less suitable for stress-relief–oriented interiors. Natural materials (e.g., wood, textiles) and indoor greenery can enhance affective regulation [29]; lighting and color also modulate psychophysiological state [30], and Wang et al. [31] found that neutral–cool correlated color temperature (~3500 K) and moderate illuminance (500–750 lx) help reduce visual fatigue and improve comfort. At the level of spatial configuration, modular, reconfigurable furniture systems allow occupants to adjust the layout according to situational needs, thereby increasing spatial adaptability, sense of control, and environmental mastery, and reducing stress caused by spatial constraints [32]. Such adaptable systems also strengthen personalization, enhancing psychological belonging and well-being. In home-office settings, highly adaptable workspaces have been shown to improve productivity and reduce stress associated with spatial limitations [33].

From a multisensory design perspective, recent research on indoor soundscapes has shifted from a narrow focus on noise control to the fit between sound and activities in different interior settings. Studies in open-plan offices show that indoor soundscapes can be described along a comfort–content dimension: when speech privacy is secured, a moderate level of conversational and background sounds can enhance environmental preference and self-rated productivity, and striving for absolute silence is not necessarily optimal [34]. In commercial and public spaces, spatial layout, brightness, thermal–humidity conditions, and visual complexity affect perceived loudness, acoustic comfort, and judgments of pleasantness and enclosure, indicating a close link between visual input and soundscape appraisal [35,36]. Survey studies on dwellings and home-working environments further suggest that the home soundscape can disrupt concentration and constrain vocal expression through traffic and neighbour noise, but can also enhance focus, relaxation and intimacy through natural sounds and music; residents actively reshape the home soundscape by opening or closing windows, playing music or using headphones [37]. Experimental work shows that combined visual, auditory and even olfactory stimuli in homes or break spaces promote recovery in indicators such as heart rate variability, and that sensory-enriched break environments are more effective for mood repair and fatigue reduction than standard settings [38,39]. In residential contexts, there is also a significant interaction between traffic noise and window views: natural views do not necessarily yield a more pleasant indoor soundscape under high traffic noise, and may even accentuate perceived incongruity [36,40]. Overall, indoor soundscape quality depends not only on sound pressure level and source type, but also on spatial form, visual environment, multisensory congruence, task type and activity demands; sound is increasingly regarded as an environmental resource that can support work performance, customer experience and everyday stress recovery.

Apart from environmental characteristics, individual-related factors, especially gender, will affect people's evaluation of soundscapes and the role of soundscapes in recovery [41,42]. Surveys show that mental health and demographic characteristics jointly mediate the pleasure and eventuality of soundscapes. Women have higher noise sensitivity and give lower pleasant evaluations in traffic or high-eventuality soundscapes [43,44]. Experimental studies have found that women have more positive evaluations of pleasant sounds such as natural sounds or music, but when exposed to noise at high volumes or under high workloads, they experience stronger feelings of irritability and fatigue, and exhibit more obvious autonomic reactions [45,46]. However, some studies suggest that demographic characteristics do not lead to significant changes in perceptual responses [47,48]. This controversy may stem from research methods, material selection, sample limitations, or overly detailed models of soundscapes' influence, making the results difficult to be universally applicable. For instance, Tarlao et al. [49] pointed out that women spend more time in public spaces and have a better experience of the sound environment, but the results may reverse in unsafe public spaces. These seemingly contradictory results point to a multi-level, detailed model of soundscapes' influence, where gender, psychological state, and social-spatial context jointly determine who benefits from which soundscapes in which spaces. Therefore, it is necessary to conduct controlled experiments in indoor, family-like environments to examine the moderating effect of gender on the recovery effect of soundscapes.

Current research on restorative soundscapes primarily focuses on urban parks and natural outdoor settings, whereas evidence on how indoor home soundscapes affect the human physiological state remains limited. This study investigates the restorative effects of home soundscapes to inform sound design in home environments. From both psychological and physiological perspectives, we analyze how soundscape exposure supports stress recovery, and how this process is modulated by home environment type and gender. Specifically, this study addresses the following questions:

(1) How do different sound types affect physiological and psychological stress recovery?
(2) How do different home environment types affect physiological and psychological stress recovery?
(3) How do sound types and home environments interact to shape recovery patterns?
(4) How does gender modulate physiological recovery responses to different soundscapes?

2. Methods

2.1. Participants

The participants were recruited through online channels (such as the school's email list and the experimental announcement platform). The recruitment process was based on the pre-set power analysis conducted in G*Power [50] ($\alpha = 0.05$, $\beta = 0.80$, medium to high effect size based on ANOVA: Repeated measures, within factors), with at least 12 individuals per group. Previous studies on the credibility of audiovisual stimuli also used similar numbers of participants [47,51]. Based on the above results, this study recruited 42 participants. To counterbalance potential order effects across the nine experimental conditions, participants were divided into three groups of 14 using a Latin square design (Table 1). This grouping ensured that each participant experienced only three conditions, preventing sensory fatigue while maintaining statistical power. There were 21 male and 21 female participants, all aged 20–27 years. The number of males and females in each group was equal (7 males and 7 females in each group). To ensure the sampling rate and data validity of the experiment, the participants were required to: have no astigmatism, be non-color blind or color deficient, have no significant difference in left and right vision; have no physical or cognitive impairments; have normal hearing, and state before the experiment that they had no hearing abnormalities and did not take psychotropic drugs. All the participants clearly read and signed the consent form before participating in the experiment.

2.2. Experimental preparation

2.2.1. Material

Following previous literature, three auditory intervention types were selected as experimental interventions: natural, urban, and music. Natural and urban stimuli were selected from the Open Science Framework (https://osf.io/kjuzr), including the research of The Sound of Nature: Examining the Influence of Natural vs. Urban Soundscapes on Connectedness to Nature (Jack Volk and Sinéad Sheehan, May 28, 2025), Passive Wildlife Audio Recordings in the Nashville Area (Nicole Creanza, Feb 22, 2023), and Open-office Noise and Information Processing (Lewend Mayiwar, Jul 28, 2023). Musical stimuli were taken from the Music Stimuli Set for an Interaction Approach to Emotional Expression through Musical Cues (Annaliese Micallef Grimaud and Tuomas Eerola, 2021) hosted on OSF. The audio was selected from research that is licensed under CC-BY Attribution 4.0 International.

Natural sounds and music were treated as pleasant sounds, whereas urban sounds were treated as unpleasant sounds. This a priori classification was based on the dominant acoustic content of each sound set and on previous findings. In general, natural sounds and slow, lyric-free music are typically experienced as pleasant and restorative at the group level [52,53], while traffic and mechanical noise are usually evaluated as unpleasant [54]. Importantly, pleasantness in the present study was ultimately assessed through participants' subjective ratings of comfort, pleasure, and related dimensions for each condition, instead of relying solely on this binary label. Natural sounds mainly consisted of birdsong, flowing water, wind, and insect calls. Urban sounds were composed of typical urban sources, including road traffic, passing vehicles, mechanical equipment, office devices, and diffuse conversational noise. In these clips, human voices appeared only as indistinct background crowd murmur without intelligible speech content, in order to avoid emotional information in speech confounding the judgment of pleasantness. Music sounds consisted of slow, continuous, lyric-free classical music excerpts with a calm character [55]. For each soundscape type, three sub-clips were prepared, each 20 s in duration. Studies have shown that the strongest response to sound occurs within one minute, and it is assumed that a 60 s stimulation duration provides the best recovery [56]. The sound pressure level was uniformly adjusted to an appropriate listening range.

To construct the visual intervention type, we first conducted an exploratory market research on mainstream video platforms (Bilibili) and residential design databases (ArchDaily). From September 15th to September 20th, 2024, we collected 94 full-space walkthrough videos of residential interiors with clear knowledge-sharing licenses, covering various elements such as spatial layout, furniture configuration, color matching, and lighting design. All selected videos were appropriately labeled and complied with copyright regulations. These videos were initially classified based on spatial type, decoration style, and image quality, and then consolidated into three categories—Capable-oriented home, Aesthetic-oriented home, and Relaxed-oriented home—defined respectively by efficient, high-utilization layouts with multi-functional furniture; visually striking, art-focused compositions with high-end materials and accent colours; and comfort-oriented, biophilic layouts with soft partitions and warm earth tones that support relaxation and recovery. We then recruited five graduate students in interior design and seven in furniture design and engineering to rate and screen the clips. Rating followed a 5-point Likert scale (1 = least consistent, 5 = most consistent) using two criteria: (1) the clip should exemplify the defining characteristics of a particular restorative home environment type; (2) the clip should express broadly relaxed-oriented qualities consistent with healing-oriented residential spaces. For each candidate clip, we computed the mean rating across all 12 students. Clips with an average score below 3.0 were rejected. Among the remaining clips, the clip with the highest mean score within each home type was retained as the final visual stimulus. The final visual stimulus set represents different types of family environments (including differences in layout, material palette, color, and spatial cues), as shown in Appendix A.

2.2.2. Experimental environment

Experiments were carried out in a controlled laboratory at Nanjing Forestry University. The experiment was conducted in a quiet laboratory located in the interior of the building. Before each experimental session,

Table 1
Experimental design group planning.

Auditory intervention types	Experimental group		
	Group1 (n = 14)	Group2 (n = 14) Visual intervention types	Group3 (n = 14)
Nature (N)	Capable-oriented home (C)	Aesthetic-oriented home (A)	Relaxed-oriented home (R)
Urban (U)	Aesthetic-oriented home (A)	Relaxed-oriented home (R)	Capable-oriented home (C)
Music (M)	Relaxed-oriented home (R)	Capable-oriented home (C)	Aesthetic-oriented home (A)

the background A-weighted sound pressure level in the empty room was measured with a sound level meter and confirmed to be below 35 dB(A). During the experiment, doors and windows were kept closed and non-essential electrical devices were switched off to maintain a stable, low-noise environment and to minimize external acoustic interference. The laboratory had plain white walls, no windows, and uniform lighting (~500 lx, ~4000 K correlated color temperature) to ensure a stable visual baseline and minimize visual distraction. Ambient conditions were kept at 24 °C and 50%–60% relative humidity.

2.2.3. Experimental equipment

This study employed the following instruments: (1) a Bitbrain wireless water-electrode EEG system, capable of recording 8–64 EEG channels (16 channels used here) at 256 Hz per channel with ±100 mV amplitude range; (2) an ErgoLAB wireless ECG sensor (±1500 μV range; 0.183 μV precision); and (3) an ErgoLAB wireless EDA sensor, sampling at 2048 Hz with 16-bit resolution and an effective range of 0–30 μS at 0.01 μS precision. (4) The HP Gaming Laptop 16-rOxxx is equipped with the ErgoLAB human–machine environment synchronization cloud platform.

2.3. Experimental procedure

This study was approved by the Experimental Animal Welfare and Ethics Committee of Nanjing Forestry University (approval No. 2024–12-20–08) and conducted in accordance with the Declaration of Helsinki. The experimental session proceeded as follows:

(1) Briefing and ethics confirmation. The experimenter explained the study goals and procedures, confirmed ethical approval, and obtained informed consent.
(2) Equipment setup and baseline recording. Participants were fitted with physiological monitoring devices. The wireless EEG system was calibrated, and baseline demographic/health information was collected.
(3) Baseline phase. Participants underwent 2 min of eyes-closed rest to achieve physiological stabilization. During this period, electroencephalogram (EEG), electrodermal activity (EDA), and electrocardiogram (ECG) signals were recorded as baseline values.
(4) Stress induction phase. Acute stress was induced using the Paced Auditory Serial Addition Test (PASAT), a standardized cognitive stressor. Time pressure was imposed to reliably provoke psychological stress. Immediately after task completion, physiological signals were captured as the pre-intervention time point.
(5) Recovery phase. Participants then wore headphones and viewed/listened to the assigned audio–visual stimulus (a combined home environment video with the corresponding soundscape) for 60 s [41]. After each exposure, they completed a subjective rating battery. A 60 s rest interval followed to prevent audio–visual fatigue. The stress task and recovery exposure cycle was repeated for each assigned condition set, for a total of three repetitions per participant. Throughout all phases, wireless sensors tracked physiological changes. The laboratory remained quiet and interruption-free; participants could withdraw at any time if discomfort occurred.
(6) Session completion. After the final exposure, all sensors were removed. The complete measurement sequence is illustrated in Fig. 1, including the extraction of pre and post-intervention differences (Δ) in physiological indices and the corresponding subjective scores.

2.4. Measurements

2.4.1. Physiological measures

Electroencephalography (EEG) is a noninvasive, cost-effective method for capturing cortical neural activity related to cognition, affect, and psychophysiological state [57–59]. The power of the θ, α, β, and γ bands is closely tied to mental state: θ activity is associated with focused attention [60]; α activity is most prominent during calm, relaxed states [61]; β activity increases with cortical arousal and alertness [62]; γ activity is linked to sustained cognitive engagement [63].

Electrocardiography (ECG) was used to obtain heart rate variability (HRV), a widely used index of autonomic nervous system (ANS) function and an objective marker of psychological stress and subsequent recovery [64,65]. HRV is derived from beat-to-beat interval variability, commonly quantified using the standard deviation of the NN (R-R) intervals (SDNN, ms), root mean square of successive differences (RMSSD, ms), the proportion of NN50 divided by the total number of NN (R-R) intervals (PNN50, %), the proportion of NN20 divided by the total number of NN (R-R) intervals (PNN20, %) and the power ratio in the low frequency and high frequency (LF/HF). During stress relief,

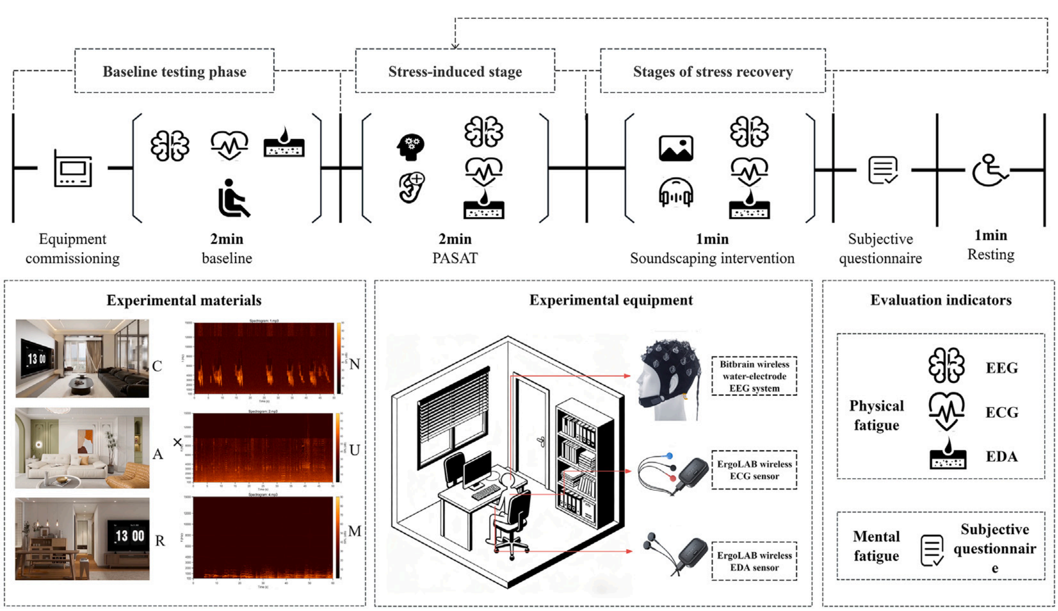

Fig. 1. Experimental procedure.

SDNN, RMSSD, PNN50, and PNN20 typically increase, whereas LF/HF decreases [66–68], indicating a shift from sympathetic dominance toward parasympathetic activation [69,70]. Electrodermal activity (EDA) provides an additional physiological index of stress, reflecting sweat gland activity and thus sympathetic nervous system activation [71]. The skin conductance level (SCL) is a standard EDA metric. Prior studies indicate that SCL is inversely related to stress recovery [72]: as stress decreases, SCL typically declines [66].

Based on current research, we believe that increases in α, θ, γ, SDNN, RMSSD, PNN50, and PNN20, as well as decreases in β, SCL, and LF/HF, can be regarded as stimulus-induced relaxation.

2.4.2. Subjective measures

The subjective rating questionnaire was developed by integrating the ISO/TS 12,913–2:2018 soundscape perception framework, psychoacoustic theory, and restorative environment theory. Eight dimensions were assessed (Comfort, Pleasant, Perceived loudness, Preference, Restorativeness, Roughness, Confusion, and Harmony). Comfort reflects perceived indoor environmental quality; Pleasant, Roughness, Confusion, and Harmony map directly onto the ISO soundscape descriptor vocabulary and capture the affective and textural qualities of the acoustic environment [73]. Perceived loudness indexes perceived acoustic intensity and relates to comfort and intrusiveness [74]; Preference reflects global evaluative tendency. This study additionally introduced a Restorativeness dimension to capture the stress-recovery function of the soundscape in a home context. This construct draws on the Perceived Restorativeness Scale (PRS) [75–77]. This approach preserves international comparability while explicitly targeting restorative outcomes in sound–home interactions. The questionnaire uses a 7-point Likert scale for subjective assessment. It is divided into 7 levels, namely extremely agitated, agitated, somewhat agitated, moderate, somewhat relieved, relieved, and incredibly relieved.

2.5. Data analysis

All data were statistically analyzed and visualized using SPSS 26.0/29.0 and Origin 2023. The physiological data were preprocessed and subjected to normality tests to assess the reliability of the subjective scales (Cronbach's α), and the average scores were calculated. The Shapiro–Wilk test was used. If the data were normally distributed, parametric tests were used; otherwise, non-parametric Friedman tests were used. Firstly, paired t-tests were conducted on various physiological indicators before and after the intervention under different acoustic scenarios to confirm the overall recovery effect of the acoustic scenarios. Subsequently, a repeated-measures ANOVA was used to examine the main and interaction effects of sound type and home environment type, with the changes in physiological indicators and subjective evaluation indicators before and after the intervention as the dependent variables. For variables with significant main effects or interaction effects, LSD post-hoc tests were further used to clarify the differences between groups.

In addition, using gender, home scene type, and sound type as independent variables, multivariate analysis of variance (MANOVA) was employed to examine the differences in physiological and subjective indicators between genders. Finally, Pearson's correlation analysis was used to assess the relationship between physiological and subjective evaluation indicators, and a heatmap was generated for visualization. All data results were presented as mean ± standard deviation (Mean ± SD).

3. Results

3.1. Effects of soundscape intervention on physiological indicators

Results of the paired t-tests across different sound perception conditions (Table 2) showed that both natural and musical sounds produced

Table 2
Summary of measurement variables.

Types of soundscapes	Physiologicalindicators	Types of home furnishings			Intervention period	Intervention period* home furnishings type
		A	C	R	P	
N	α	1.56±1.34***	0.5±0.82	2.05±2.17***	<0.001***	0.035*
	β	−1.34±2.02	−1.32±1.68	−0.83±0.73	<0.001***	0.629
	γ	2.82±1.12***	1.94±1.29***	1.08±0.67***	<0.001***	<0.001***
	θ	1.5±1.37	1.28±1.17	1.73±1.64	<0.001***	0.703
	ΔSCL	0.37±0.59	0.28±0.26	1.11±1.2***	<0.001***	0.015*
	ΔSDNN	16.92±4.72	13.2±9.24	12.38±7.92	<0.001***	0.302
	ΔRMSSD	−0.11±5.47	6.95±11.23	1.11±8.31	0.068	0.103
	ΔPNN50	4.88±2.17	3.54±0.97	4.49±2.95	<0.001***	0.299
	ΔPNN20	9.03±8.2***	7.57±4.39***	3.46±1.93*	<0.001***	0.041*
	ΔLF/HF	−1.39±0.41***	−1.18±0.82***	−3.54±0.95***	<0.001***	<0.001***
U	α	0.3±1.37	−0.14±1.31	0.49±1.6	0.338	0.494
	β	1.75±1.08	1.4±2.27	0.7±1.17	<0.001***	0.228
	γ	−0.93±0.92*	−2.2±1.75***	−0.55±1.06	<0.001***	0.004**
	θ	−1.39±0.86	−1.54±2.01	−0.29±1.3	<0.001***	0.06
	ΔSCL	1.87±0.91***	2.47±0.87***	−1.17±1.05***	<0.001***	<0.001***
	ΔSDNN	−4.89±7.24***	−9.25±6.85***	−12.1±6.8***	<0.001***	0.045*
	ΔRMSSD	−3.55±4.95	−1.57±5.09	−0.36±2.68	0.015*	0.199
	ΔPNN50	−4.42±5.57	−4.04±5.75	−1.78±3.73	<0.001***	0.401
	ΔPNN20	−4.88±5.17	−0.14±4.28	−1.3±5.57	0.014	0.054
	ΔLF/HF	0.96±0.7***	2.98±1.19***	1.15±0.77***	<0.001***	<0.001***
M	α	1.64±1.4	0.77±1.38	0.99±2.81	<0.001***	0.486
	β	−1.68±1.19***	0.94±0.99**	−0.77±1.22**	0.007**	<0.001***
	γ	1.98±3.52	0.15±1.03	1.21±1.2	0.002**	0.108
	θ	1.42±3.09	0.61±0.8	1.72±1.13	<0.001***	0.311
	ΔSCL	2.97±0.37***	1.95±0.42***	2.18±1.18***	<0.001***	0.002**
	ΔSDNN	6.58±4.1	5.85±2.61	4.44±2.67	<0.001***	0.23
	ΔRMSSD	4.29±2.41***	1.73±2.19*	2.84±1.77***	<0.001***	0.019*
	ΔPNN50	2.55±1.63***	1.13±1.68*	4.14±1.59***	<0.001***	<0.001***
	ΔPNN20	4.4±4.44	4.58±4.52	2.34±7.93	<0.001***	0.567
	ΔLF/HF	−0.87±0.93**	0.004±1.05	−0.77±0.64**	<0.001***	0.042*

significant improvements across most physiological indicators, whereas urban sounds were associated with adverse effects.

For electroencephalography (EEG) activity, exposure to natural sound significantly increased α, γ, and θ power and decreased β power, with the most pronounced effects observed in the relaxation-oriented and aesthetic-oriented homes. The urban sound led to a significant increase in β power and decreases in θ and γ power, while α-band power showed no significant change. The music sound significantly enhanced α, γ, and θ power and reduced β power ($p < 0.01$), and the aesthetic home exhibited the most substantial regulatory effect on β activity.

For electrodermal activity (EDA), exposure to the natural sound significantly increased skin conductance level (SCL) ($p < 0.001$), with the most significant improvement observed in the relaxation-oriented home. Under the urban sound, overall SCL tended to increase ($p < 0.001$), although in the relaxation-oriented home it decreased. The music sound produced significant elevations in SCL across all three home environment types ($p < 0.001$), with the most substantial effect in the aesthetics-oriented home.

For cardiac autonomic indices derived from heart rate variability (HRV), the natural sound significantly increased SDNN, PNN50, and PNN20, while reducing the LF/HF ratio; changes in RMSSD were not significant. The urban sound significantly reduced SDNN, RMSSD, PNN50, and PNN20, and increased LF/HF, indicating elevated sympathetic activation. The music significantly improved SDNN, RMSSD, PNN50, and PNN20 ($p < 0.001$), except in the capability-oriented home, where it also reduced LF/HF ($p < 0.01$). The aesthetic-oriented home generally showed the most significant improvement in these HRV-based indicators.

3.2. Effects of sound type in soundscapes on psychophysiological stress relief

3.2.1. Effects of soundscape type on physiological and psychological stress relief

Both subjective and objective indicators show that the results are significantly influenced by the type of sound. Comparisons across sound types are shown in Fig. 2. The EEG results show that the recovery effect of natural sound is the best, while that of urban sound is the worst (Fig. 2(a)). For the α wave: music ($p = 0.013$) and nature ($p = 0.002$) are both higher than urban sounds, and there is no significant difference between the two. For the β wave, the opposite is true, music and nature are both lower than urban sound, and the decrease in nature is greater than that in music ($p = 0.038$). For the γ wave and the θ wave, they are all nature > music > urban.

The correlation influence of the sounds on the skin conductance and electrocardiogram indicators is shown in Fig. 2(b). In terms of skin conductance activity, the ΔSCL perceived by the music sounds is significantly higher than that of the urban sound and the nature sound, and the ΔSCL of the urban sound is slightly higher than that of the nature sound

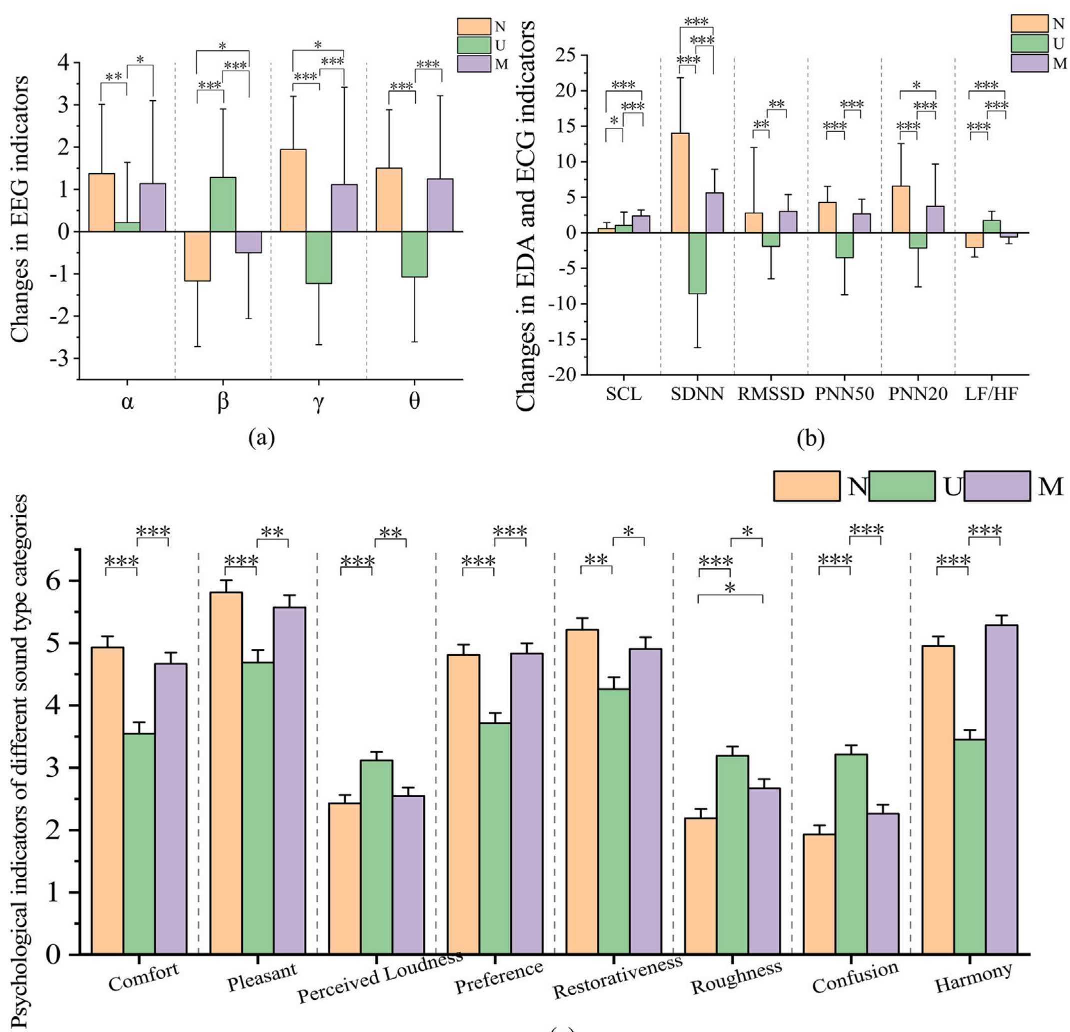

Fig. 2. The influence of sound scene type after intervention on physiological indicators: (a) EEG indicators; (b) EDA indicators and ECG indicators; (c) subjective questionnaire; (*) 0.01 < p < 0.05, (**) 0.001 < p < 0.01, (***) 0.0001 < p < 0.001, (>***) p < 0.0001.

($p=0.013$). In terms of heart rate variability, the ΔSDNN and ΔPNN20 indicators show that nature > music > urban, with significant differences between groups, indicating that natural sounds have the optimal effect on autonomic nerve balance. The ΔRMSSD and ΔPNN20 results show that natural and music sounds are significantly higher than urban sounds, and there is no significant difference between natural and music sounds. ΔLF/HF is the highest in the urban condition ($p<0.001$), and nature is significantly lower than music ($p<0.001$).

As shown in Fig. 2(c), for positive perceptual dimensions (comfort, pleasant, preference, restorativeness, and harmony), both the natural sound and the music sound received significantly higher ratings than the urban sound, with no significant difference between natural and music. For the negative perceptual dimensions (perceived loudness, roughness, and confusion), the urban sound was rated significantly higher than both natural and music, indicating greater perceived disturbance. For roughness, the natural sound was rated significantly lower than the music sound ($p = 0.026$), whereas for the remaining dimensions, no significant differences were found between natural and music.

3.2.2. Effects of home environment type in soundscapes on psychophysiological stress relief

The type of home environment also had a significant effect in each frequency band ($p = 0.034$ - 0.045), but the effect size was relatively small. In terms of the autonomic nervous system, only SDNN ($p = 0.027$, $\eta^2 = 0.063$) and SCL ($p < 0.001$, $\eta^2 = 0.228$) showed significant effects. For EEG indicators (Fig. 3(a)), the capable-oriented home showed significantly lower α band power than both the aesthetic($p = 0.032$) and the relaxed ($p = 0.030$) oriented home. In contrast, for β band power, the capable-oriented home was significantly higher than the aesthetic-oriented home ($p = 0.018$) and the relaxed-oriented home ($p = 0.046$). For the γ band, the aesthetic-oriented home was significantly higher than both the capable-oriented home ($p < 0.001$) and the relaxed-oriented home ($p = 0.046$), with no significant difference between the latter two. For the θ band, a significant difference was observed only between the capable-oriented home and the aesthetic-oriented home ($p = 0.010$).

As shown in Fig. 3(b), for ECG and EDA indicators, both the aesthetic-oriented home and the capable-oriented home exhibited significantly higher ΔSCL than the relaxed-oriented home ($p < 0.001$), with no significant difference between the two former types. For ΔSDNN, only the aesthetic-oriented home scored significantly higher than the relaxed-oriented home ($p = 0.008$), and no other pairwise comparisons were significant.

The types of homes showed significant differences across seven dimensions, except for the dimension of pleasure level, where the difference did not reach a significant level ($p = 0.116$). As shown in Fig. 3(c), different home environment types produced significant differences across multiple subjective dimensions. For the positive perceptual dimensions (comfort, preference, restorativeness, and harmony), both the aesthetic-oriented and relaxed-oriented homes scored significantly higher than the capable-oriented home. Notably, the relaxed-oriented home received significantly higher ratings for restorativeness than the aesthetic-oriented home ($p = 0.005$). For pleasant, no significant differences were found among the three home types. For the negative perceptual dimensions (perceived loudness, roughness, and confusion), the capable-oriented home scored significantly higher (i.e., worse) than the other two home types. The aesthetic-oriented home was rated

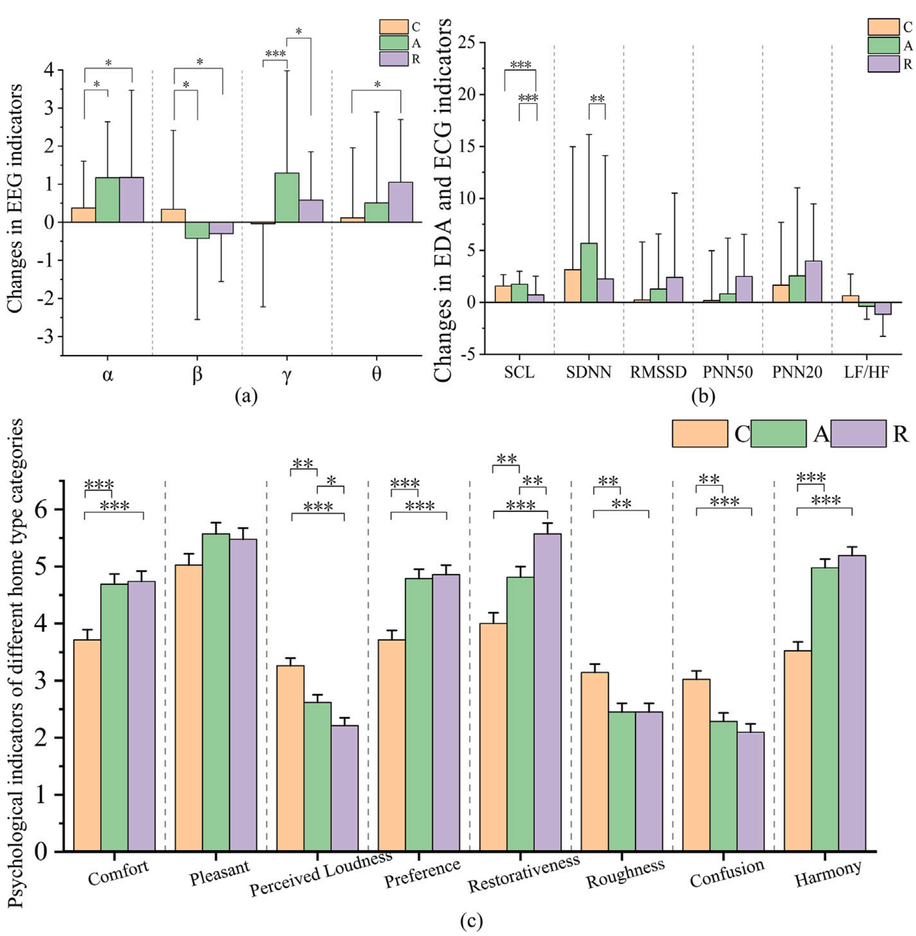

Fig. 3. The influence of home type after intervention on physiological indicators: (a) EEG indicators; (b) EDA indicators and ECG indicators; (c) subjective questionnaire; (*) $0.01 < p < 0.05$, (**) $0.001 < p < 0.01$, (***) $0.0001 < p < 0.001$, (>***) $p < 0.0001$.

significantly higher than the relaxed-oriented home in perceived loudness (p = 0.036), whereas the two did not differ significantly in roughness or confusion. Taken together, the capable-oriented home is associated with more substantial negative perceptual load, whereas the relaxed-oriented home achieves the most favorable evaluations across most positive restorative dimensions.

3.2.3. Effects of audio-visual interaction in the sound scene on psychophysiological stress relief

The interaction effect results showed that there was a significant interaction between β ($p < 0.001$) and γ waves ($p = 0.031$) in terms of sound environment and home, while there was no effect for α and θ waves; in the autonomic nervous system, the SCL interaction effect was the strongest ($p < 0.001$), and PNN20 was also significant ($p = 0.009$), while the other indicators were not significant. For the indicators with significant effects, an LSD post-hoc test was conducted to further reveal the differences between the groups. The outcomes of this analysis are illustrated in Figs. 4 and 5.

The simple effect analysis for EEG indicators (Fig. 4) showed the following patterns: For $\Delta\beta$ power, within the aesthetic-oriented home, urban sound was significantly higher than both music and natural sounds, with no significant difference between music and natural sounds. Within

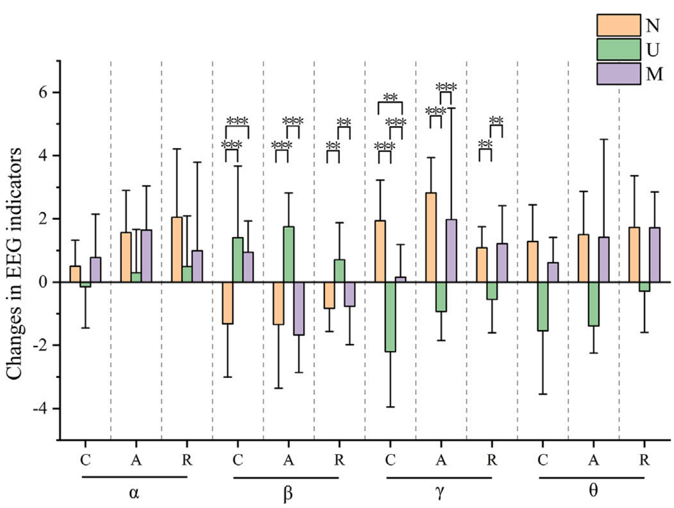

Fig. 4. EEG changes under different acoustic environment interactions; (*) 0.01 < p < 0.05, (**) 0.001 < p < 0.01, (***) 0.0001 < p < 0.001, (>***) p < 0.0001.

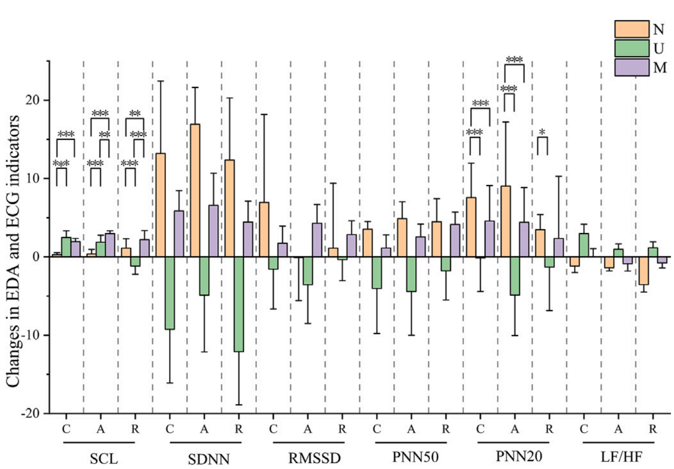

Fig. 5. EDA and ECG changes under different acoustic environment interactions; (*) 0.01 < p < 0.05, (**) 0.001 < p < 0.01, (***) 0.0001 < p < 0.001, (>***) p < 0.0001.

the capable-oriented home, the natural sound produced significantly lower $\Delta\beta$ than both urban and music ($p < 0.001$), and there was no significant difference between music and urban. Within the relaxed-oriented home, the urban sound again yielded significantly higher $\Delta\beta$ than music ($p = 0.008$) and natural ($p = 0.006$), while music and natural did not differ. For $\Delta\gamma$ power, in the aesthetic-oriented home, both music and natural sounds were significantly higher than in urban homes, with no difference between music and natural sounds. In the capable-oriented home, the natural sound was significantly higher than both music ($p = 0.004$) and urban ($p < 0.001$), and music was also higher than urban ($p < 0.001$). In the relaxed-oriented home, both music and nature again produced significantly higher $\Delta\gamma$ than in the urban home ($p = 0.005$; $p = 0.008$), and did not differ from each other.

The results of the simple effect analysis of the EDA and ECG indicators are shown in Fig. 5. Within the aesthetic-oriented home, the music sound produced significantly higher ΔSCL than both natural ($p < 0.001$) and urban ($p = 0.006$), and urban was significantly higher than natural. Within the capable-oriented home, the music sound again produced significantly higher ΔSCL than natural sound, and natural sound was significantly lower than urban sound. In the relaxed-oriented home, the urban sound produced significantly higher ΔSCL than both natural and music sounds.

For the HRV indicator PNN20, within the aesthetic-oriented home, the natural sound was significantly higher than both music and urban, with no difference between music and urban. Within the capable-oriented home, the natural sound was again significantly higher than both music and urban, and the contrast between music and urban approached significance ($p = 0.050$), suggesting that natural sound exerts the most substantial regulatory effect on autonomic balance in this setting. In the relaxed-oriented home, natural sounds were significantly higher than in urban environments ($p = 0.014$), but did not differ significantly from music. Overall, these results indicate that physiological stress modulation depends on how specific sounds are embedded within specific home environments, rather than on either factor in isolation.

The interaction effect between sound and home environment was significant in terms of comfort ($F = 3.244$, $p = 0.015$), preference ($F = 4.260$, $p = 0.003$), roughness ($F = 2.866$, $p = 0.026$), and harmony ($F = 8.092$, $p < 0.001$), but no significant interaction was observed in terms of pleasant, perceived loudness, restorativeness, and confusion. The simple effect analysis for subjective indicators (Fig. 6) showed: For comfort, both the relaxed-oriented home and the aesthetic-oriented home were rated significantly higher under the natural and music sounds than under the urban sound, with the relaxed-oriented × natural combination performing best. In contrast, within the capable-oriented home, there were no significant differences among the three sounds. For pleasant, in the capable-oriented home, both natural ($p = 0.02$) and music ($p = 0.029$) scored significantly higher than urban. Similar advantages for natural and music over urban were observed in the aesthetic-oriented and relaxed-oriented homes, with the relaxed-oriented × natural pairing again yielding the highest ratings. For perceived loudness, ratings were generally higher (i.e., worse) in the capable-oriented home, with the functional-efficiency × urban combination showing the most unfavorable outcome.

For preference, both the relaxed-oriented home and the aesthetic-oriented home, under natural and music sounds, were rated significantly higher than under urban sound ($p < 0.001$). Within these, the relaxed-oriented × natural and relaxed-oriented × music combinations were most preferred, whereas in the capable-oriented home, differences among sounds were not significant. For restorativeness, the relaxed-oriented × music condition achieved the highest score (6.071), while the capable × urban combination was the lowest (3.357). Across all three home types, the natural sound consistently received the highest ratings, underscoring its restorative advantage. For the negative perceptual dimensions—roughness and confusion—the capable × urban pairing produced significantly higher scores than all other combinations, indicating the strongest negative perceptual impact. In contrast, both

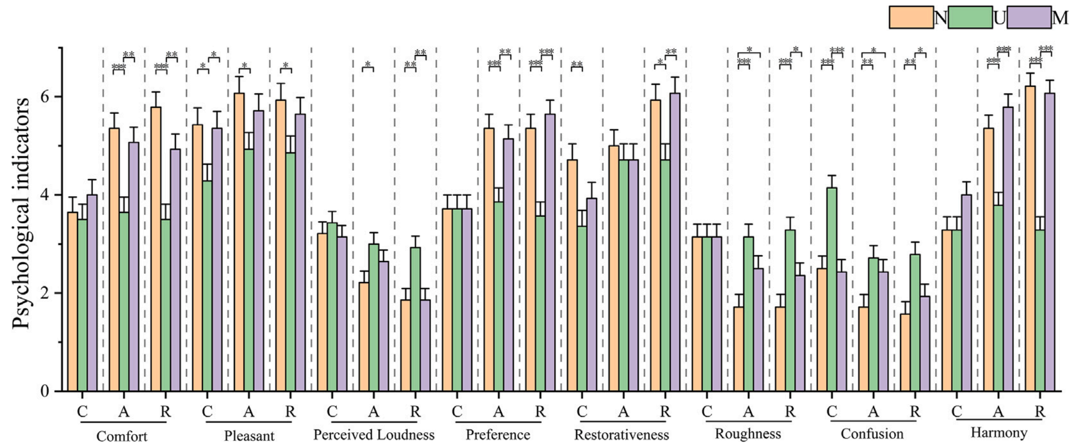

Fig. 6. Subjective indicators changes under different acoustic environment interactions; (*) 0.01 < p < 0.05, (**) 0.001 < p < 0.01, (***) 0.0001 < p < 0.001, (>***) p < 0.0001.

the aesthetic-oriented and relaxed-oriented homes paired with natural or music yielded the lowest scores. For harmony, the aesthetic-oriented and relaxed-oriented homes under natural and musical conditions produced the highest ratings, significantly better than those of urban homes. In the capable-oriented home, music was rated highest, and urban was rated lowest.

3.3. Gender-related group differences

A multivariate analysis of variance (MANOVA) was conducted with gender (male, female), home environment type (three levels), and sound perception (three levels) as independent variables, and both physiological and subjective indicators as dependent variables. The results showed significant main effects of gender on $\Delta\beta$ power ($p = 0.025$) and ΔSCL ($p = 0.007$). A significant gender $\times$ sound interaction emerged for $\Delta\gamma$ power ($p = 0.038$). In addition, a significant three-way interaction of gender $\times$ home environment type $\times$ sound type was observed for $\Delta\theta$ power ($p = 0.012$) and ΔSCL ($p < 0.001$). By contrast, gender did not exert a significant effect on any of the subjective rating dimensions.

(1) EEG indicators Post hoc comparisons showed that females exhibited significantly higher $\Delta\beta$ power than males ($p = 0.025$). For $\Delta\gamma$ power, both males and females displayed the same trend: the natural sound and the music sound produced significantly higher values than the urban sound (Fig. 7), and there was no significant difference between natural and music. Females, however, were more sensitive to the contrasts between natural-urban ($p < 0.001$) and music-urban ($p < 0.001$). For $\Delta\theta$ power (Fig. 8), females showed a consistent pattern across all home environment types, with natural > music > urban, and all differences were significant ($p < 0.05$). Male participants showed significant differences only in the aesthetic-oriented home, where both natural sound ($p = 0.001$) and music sound ($p < 0.01$) were higher than urban sound. Overall, females demonstrated more pronounced differentiation in θ band responses.

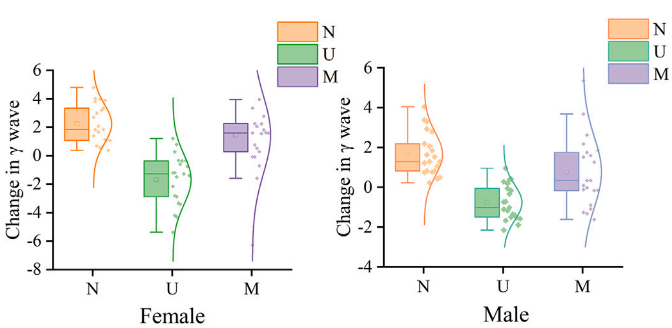

Fig. 7. The results of pairwise comparison of brain wave changes.

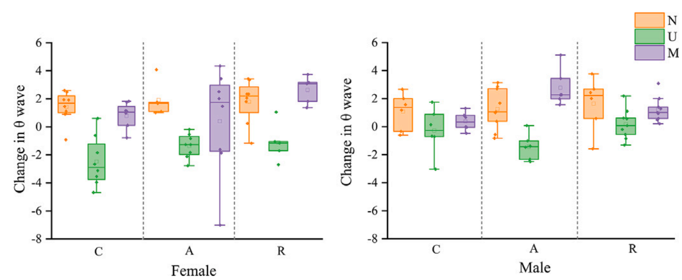

Fig. 8. The results of pairwise comparison of brain wave changes.

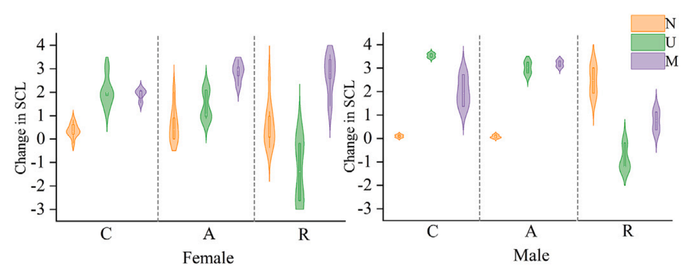

Fig. 9. The results of pairwise comparison of EDA changes.

(2) Electrodermal and cardiac autonomic indicators Post hoc testing on the main effect of gender revealed that males showed significantly higher ΔSCL than females ($p = 0.007 < 0.05$). Further analysis of the sound $\times$ home environment type $\times$ gender interaction (Fig. 9) showed that, in the capability-oriented home, female participants exhibited the highest ΔSCL to the music sound. Significant pairwise differences appeared between natural - music ($p < 0.001$) and urban - music ($p < 0.001$). Male participants, in contrast, showed significantly lower SCL under the natural sound than under both the urban sound ($p < 0.001$) and the music sound ($p < 0.001$), with no significant difference between urban and music ($p > 0.05$). In the relaxation-oriented home, female participants showed higher ΔSCL to the natural sound than to the urban sound ($p < 0.001$), but significantly lower than to the music sound

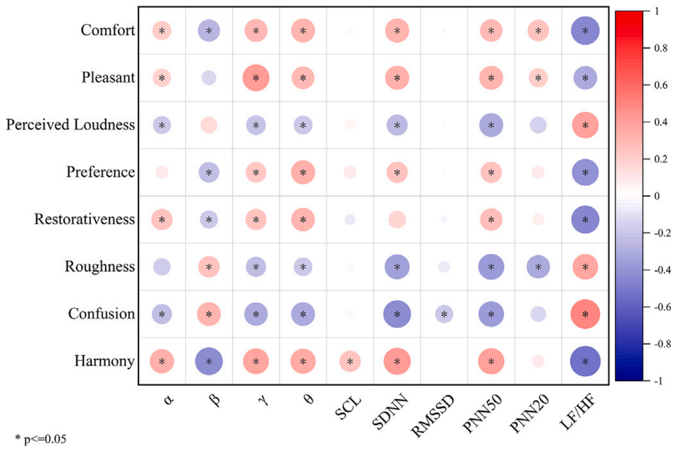

Fig. 10. Results of correlation analysis between physiological index and subjective index.

(p < 0.001). Male participants showed the strongest sympathetic activation to natural sounds, with a mean ranking of urban < music < natural; the contrast between natural and music was also significant (p < 0.001).

3.4. Correlation analysis between physiological indicators and subjective indicators

Previous work by Erfanian et al. [78] has shown that physiological monitoring techniques and psychological self-report measures are complementary.

The correlation heatmap (Fig. 10) demonstrated robust associations between subjective ratings and physiological indicators. Specifically, the positive subjective dimensions — comfort, pleasant, preference, restorativeness, and harmony — showed significant positive correlations with EEG α, θ, and γ band power and with heart rate variability (HRV) indices including SDNN, PNN50, and PNN20. This indicates that stronger positive experiential states are accompanied by greater physiological relaxation and enhanced restorative responses. In contrast, the negative subjective dimensions-perceived loudness, roughness, and confusion — were significantly negatively correlated with EEG α, θ, and γ band power and with the restorative HRV indices (SDNN, PNN50, PNN20), and were significantly positively correlated with β band power and the LF/HF ratio. This pattern suggests that more negative perceptual evaluations are accompanied by stronger sympathetic activation and elevated stress-related arousal.

Overall, the subjective positive experiences are highly consistent with parasympathetic nerve activity and restorative brainwave patterns, while negative experiences align with a sympathetic nerve-dominated state of tension. This indicates good consistency and complementarity between subjective perceptions and physiological responses, providing a basis for future multidimensional evaluation of the restorative environment.

4. Discussion

4.1. Restorative effects of different soundscape types on psychophysiological stress

The results of this study showed that, under natural and musical soundscapes, EEG α, θ and γ band power increased, β band power decreased, and skin conductance levels were significantly lower than in the urban sound, heart rate variability (HRV) indices (SDNN, RMSSD, PNN50, PNN20) increased, and the LF/HF ratio decreased. In contrast, under the urban sound, α power was significantly lower than under natural and musical sounds, θ and γ power decreased markedly, β power increased, HRV indices decreased, and LF/HF increased. This pattern is consistent with a large body of literature. Relaxed or low-pressure contexts are typically accompanied by increased α and θ and decreased β [63,79,80]. Natural or biophilic soundscapes, compared with urban sounds, tend to increase SDNN, RMSSD, PNN50 and PNN20 [66,81–83] and reduce sympathetic dominance, as reflected in lower LF/HF [84–86]. Taken together, these findings indicate that natural and musical sounds can effectively promote activation of the parasympathetic nervous system, reduce individual tension, and thus relieve stress, with the cortical state shifting from a "tense–alert" mode to a "relaxed–engaged" mode. In contrast, the sounds of the city exhibit a typical state of sympathetic nervous system excitation - heightened alertness, increased tension, and increased emotional burden. Subjective evaluations followed the same pattern: natural and musical sounds were rated substantially higher than the urban sound in comfort, pleasure, preference, restorativeness, and harmony, whereas the urban sound received the highest scores for perceived loudness, roughness, and confusion, implying that it is experienced as "noisy, intrusive, and difficult to relax in."

It is noteworthy that, although both natural and musical sounds are restorative, they may operate through different regulatory mechanisms. The natural sound produced larger improvements in ΔSDNN, PNN20, and LF/HF, indicating stronger promotion of autonomic balance and a more "calming/soothing recovery" profile [20]. The musical sound also improved recovery-related indices, but it was often accompanied by elevated skin conductance level (SCL), especially in aesthetically oriented interiors. This pattern suggests that music can induce positive arousal—a state in which participants report pleasure and a sense of being "healed," yet exhibit moderate physiological activation rather than purely passive relaxation [87]. The correlation between subjectivity and objectivity also indicates that SCL and harmony are positively correlated. This phenomenon is similar to the findings of Koivisto et al [88]. In their experiment, participants' HRV (including SDNN) increased significantly when they imagined natural landscapes, and skin conductance (SCL) rose simultaneously.

4.2. Restorative effects of different residential setting types on psychophysiological stress

The advantage of the Relaxed-oriented home stems from the joint effects of a warm, neutral palette and high-tactility, biophilic materials (wood, textiles, linen). Warm neutral colors reduce defensive attention and tension, while the tactile and acoustic properties of wood/textiles jointly promote parasympathetic rebound, yielding a stable relaxation pattern in EEG and HRV that aligns with our findings [89,90]. Subjectively, Comfort, Pleasant, Harmony, and Restorativeness increase concurrently. Bower et al. [91] also reported greater theta power over frontal and left temporal regions in interiors rated higher on self-reports, supporting the view that a comfortable domestic visual environment can modulate stress-related neural activity. Consistently, SCL in the Relaxed-oriented home is significantly lower than in the Aesthetic-oriented and Capable-oriented homes. The introduction of soft partitions (e.g., carpets and indoor plants) and natural elements (e.g., unfinished wood and linen) creates a biophilic, soothing ambiance that effectively reduces sympathetic activation and facilitates relaxation. Prior studies further show that vegetation-rich, comfortable residential settings reduce SCL and enhance restorative perceptions [92,93], a pattern consistent with the color-valence effect and the dual gains of materiality.

The Aesthetic-oriented home exhibits a co-occurrence of recovery and positive arousal, suggesting that visual aesthetics can alleviate stress and support autonomic balance [94]. SCL is higher than in the Relaxed-oriented home, indicating some sympathetic activation. Increasing visual complexity and aesthetic quality elevates Preference-consistent with the "positive arousal" model [95–97], in which favorable emotional arousal and preference rise, and SCL may increase even when the bodily state is relaxed, reflecting emotional attention and sensory engagement [88]. Similar findings were reported by Grassini et al., who found that SCL during nature-video viewing exceeded that during urban-video viewing [98]. On the subjective scales, the Aesthetic-oriented home scores well on Comfort, Preference, and Harmony, but shows higher Perceived loudness than the Relaxed-oriented home, plausibly due to its emphasis on visual impact and material complexity.

By contrast, the Capability-oriented home shows insufficient relaxation and relatively higher arousal: increases in α, θ, γ power are limited, β power is higher, and the lowest scores appear on Comfort, Pleasant, Harmony, and Restorativeness, while Perceived Loudness, Roughness, and Confusion are highest. This pattern is consistent with the combination of cold neutral color blocks and a high proportion of highly reflective artificial materials, such as glass/resin—overuse of neutrals can induce monotony and delay recovery [97]. Artificial materials are inferior to wood in stress alleviation [99]. Moreover, interiors optimized excessively for efficiency and function often lack sensory support, thereby increasing psychological strain and physiological arousal [100].

4.3. Interactive effects of soundscape type and home environment type on psychophysiological stress recovery

In the urban acoustic environment, all three types of homes have adverse effects. The β wave increase in the comfortable home is less than that in the aesthetic and capable homes, and the ΔSCL is the lowest, which can buffer neural tension and sympathetic activation. Under the natural sound, the aesthetic home shows significant increases in γ waves and SDNN, with strong cognitive and autonomic nerve activation, but the ΔSCL is higher, which is prone to trigger “positive arousal”; in the relaxed-oriented home, the γ wave increase is slight, and the ΔSCL is the lowest, showing more relaxation. Under the music acoustic environment, the subjective comfort and restorativeness degree scores of the relaxed-oriented home are the highest, while the aesthetic home has better pleasant and harmony, and the capable-oriented home has the weakest indicators overall. In the relaxed-oriented home, both the natural and music sounds significantly reduce β waves, increase γ waves, and improve HRV, outperforming the urban sound; the inhibition of ΔSCL by the natural sound is particularly prominent. In the aesthetic home, the natural acoustic environment significantly increases ΔPNN20 and γ waves, and subjective pleasure is excellent; both are significantly higher than in urban environments. In the capable-oriented home, the natural sound significantly improves HRV and reduces β waves, with a compensatory effect, but the recovery potential is lower than that of the relaxed and aesthetic homes.

Taken together, the findings suggest that sound type exerts a more decisive influence than home environment type. Natural and music sounds consistently showed clear advantages in EEG (reduced β power, increased γ power), HRV (enhanced SDNN, RMSSD, PNN20), and subjective restoration scores, whereas the urban sound generally produced adverse effects. This trend remained stable across all three home environments, implying that sound is the primary driver of restorative outcomes. By contrast, the home environment mainly acts as a moderator: the relaxation-oriented setting buffers the adverse impact of urban sound; the aesthetics-oriented setting amplifies restorative benefits from natural sound, albeit with positive arousal; and the capability-oriented setting shows the weakest inherent restorative capacity. In this sense, within indoor restorative sound interventions, the sound type sets the baseline level of recovery, while spatial form shapes whether that recovery is amplified or attenuated.

Although Jeon et al. [47] reported that visual and auditory information contributed 76% and 24%, respectively, to overall environmental satisfaction—thereby underscoring the dominance of visual factors in urban audio-visual design that conclusion was primarily derived from subjective preference ratings and often contrasted with highly idealized “natural vs. urban” visuals. By contrast, Van et al. [24] showed that adding natural sound to an otherwise urban visual scene can still promote relaxation, even in the absence of visual nature [24]. In light of the present results, sound should not be treated as a secondary background variable in residential restorative design. Instead, it should be regarded as an active design lever on par with visual form: pairing natural or musical soundscapes with a relaxation-oriented setting can further enhance calmness and perceived healing [101]; matching natural soundscapes with an aesthetics-oriented setting can amplify positive arousal; and in capability-oriented settings, deliberate soundscape intervention may be necessary to prevent sustained high-pressure states.

4.4. Gender differences in soundscape-induced physiological recovery

This study shows that gender significantly moderates the restorative effects of different soundscapes on specific physiological indicators. Specifically, females exhibited significantly higher $\Delta\beta$ wave power than males, suggesting greater tension and cognitive load under stress. However, for $\Delta\gamma$ and $\Delta\theta$ wave activity, females were more sensitive to the contrast between natural/music sounds and urban sounds, showing stronger recovery effects. Male EEG responses were more stable, and a clear restorative advantage of natural and musical sounds was observed, mainly within the aesthetics-oriented home environment. This indicates that females are generally more responsive to the restorative benefits of positive soundscapes, whereas males tend to exhibit flatter responses across contexts. A similar trend was reported by Li & Kang [102], who found gender differences in EEG-based emotional responses, with females showing larger emotion-related EEG changes.

For autonomic nervous system (ANS) measures, males showed significantly higher ΔSCL than females, suggesting stronger sympathetic skin conductance responses during both stress and recovery phases [103,104]. Prior studies similarly indicate that females often exhibit relatively greater parasympathetic activity [105–107], consistent with our findings. This gender difference also manifested differently across home environments: Across the three home types, females showed the highest ΔSCL under the music sound, suggesting a stronger positive arousal effect of music for women. In the capable-oriented home, males showed significantly lower ΔSCL to natural sound than to urban or music sounds ($p < 0.001$), indicating suppressed sympathetic activation to natural sound. In the relaxed-oriented home, males exhibited the ordering urban < music < natural, meaning that natural sounds evoked the strongest sympathetic activation, suggesting a recovery mode that is more arousal-driven than purely calming. Importantly, no significant gender differences emerged in subjective ratings, suggesting that even when males and females differ physiologically, their perceived restorative experience can remain similar.

Overall, females demonstrated a more robust and consistent recovery pattern under the natural sound, characterized by clearer differences in θ wave and γ wave activity and relatively lower sympathetic activation as reflected in SCL, suggesting stronger restorative capacity. Under the musical sound, the SCL of women also showed significant changes, which might be caused by adaptive stress responses (moderate sympathetic nerve activation) [108] and positive preference arousal [88]. By contrast, males’ recovery responses were more contingent on the spatial context: in the capability-oriented home, the natural sound suppressed sympathetic activation, whereas in the relaxation-oriented home, the same natural sound elicited heightened sympathetic activation.

4.5. Limitations and future directions

This study systematically evaluated stress recovery along three dimensions—sound type, home environment type, and gender differences—but several limitations remain. Firstly, the sample mainly consisted of young people, and participants with known cognitive impairments were not included. This limits the general applicability of the research results to the aging population and those with diverse cognitive conditions. Second, the experiment was conducted in a controlled setting rather than in actual lived environments, which restricts ecological validity. Third, our indicators focused on EEG, HRV, and SCL, without incorporating additional multimodal biomarkers such as cortisol or functional brain connectivity, so the underlying mechanisms are not yet fully elucidated. Moreover, the range of sounds and interior typologies was necessarily constrained and did not explicitly account for individual preference patterns or more complex composite scenes, which may lead to an underestimation of personalization effects. Additionally, although we followed a structured analysis pipeline, we did not apply formal family-wise error corrections for multiple paired t-tests and LSD post-hoc comparisons. In subsequent studies, multiple comparison corrections such as Bonferroni should be applied to further validate the reliability of the findings. Finally, exposure was short-term, leaving the long-term impact of repeated or chronic restorative soundscape interventions untested.

Future work can advance in several directions: (1) broaden the sample to include diverse ages, occupations, and cultural backgrounds, explicitly modeling individual differences to improve external validity; (2) conduct studies in real homes or immersive virtual home environments to enhance ecological realism; (3) integrate multimodal physiological markers (e.g., endocrine measures, functional brain connectivity) together with behavioral data to clarify recovery mechanisms; (4) develop multisensory, personalized intervention strategies tailored to specific groups (e.g., women, older adults, chronically stressed populations); and (5) perform longitudinal and cross-context studies to assess the durability and transferability of soundscape-based restorative interventions.

Beyond architectural and laboratory applications, the present findings have important implications for technology-mediated environments. As controlled auditory and audiovisual stimuli are increasingly delivered through virtual reality (VR) systems, immersive games, and digital therapeutic platforms, the observed advantages of natural and musical soundscapes in restorative home settings can be translated into concrete design principles. (1) Restorative VR experiences could embed natural sounds and visually relaxing home-like scenes to provide short “micro-break” episodes during cognitively demanding work. (2) Soundscapes for immersive video gaming could be optimized to balance arousal and recovery, for example by integrating natural or music-based sound layers into menus, loading screens, or cool-down phases. (3) VR-based stress recovery interventions may use our psychophysiological indicators as outcome measures to fine-tune multisensory environments. (4) Digital wellbeing applications could incorporate adaptive sound and sound–space engagement systems that dynamically adjust sound type and intensity according to the user’s current stress state and contextual demands.

5. Conclusions

This study experimentally demonstrated the significant roles of soundscape type, residential environment, and gender differences in stress recovery, and further revealed the complementary relationship between physiological indicators and subjective evaluations. Specifically:

(1) The effect of soundscape type is pronounced. Both the natural soundscape and the music soundscape showed clear restorative advantages, including enhanced EEG α, θ, and γ band power, reduced β band power, improved HRV indices, and higher subjective ratings of comfort, pleasant, and related dimensions. In contrast, the urban soundscape generally elicited stronger sympathetic nervous activation and a heightened state of tension. Notably, the natural soundscape was particularly effective in regulating the autonomic nervous system (ANS).

(2) The home environment type functions as a moderator of restorative outcomes. The relaxed home environment performed best in both subjective emotional regulation and physiological relaxation. The aesthetics-oriented home environment tended to induce positive arousal, characterized by elevated SCL, yet it also showed advantages in cognitive engagement and autonomic balance. The capable-oriented home environment demonstrated the weakest inherent restorative potential and therefore requires compensatory support from the sound.

(3) There is a significant interaction between sound type and home environment type. The sound serves as the primary driver of the restorative effect, while the home environment modulates this effect. A relaxation-oriented home environment can buffer the negative impact of urban sound; an aesthetics-oriented home environment achieves maximal restorative benefit under natural sound; and a capability-oriented home environment shows limited overall restorative capacity.

(4) Gender differences are evident. Female participants were more sensitive to the restorative influence of natural and musical soundscapes, exhibiting stronger EEG response changes and lower levels of sympathetic activation. Male participants showed recovery patterns that were more dependent on the specific spatial context, and tended to exhibit stronger skin conductance responses.

Taken together, these findings indicate that the soundscape in the home environment should not be regarded merely as a passive target of noise control, but as an active resource for promoting recovery. Future restorative residential design and health-oriented living environments should prioritize the intentional integration of natural and musical sounds, aligning them with the comfort qualities and aesthetic characteristics of the interior to achieve multisensory optimization. Furthermore, longitudinal studies in real-world living contexts and across different age groups are needed to enhance the ecological validity and practical applicability of soundscape-based interventions.

CRediT authorship contribution statement

Chengmin Zhou: Writing – review & editing, Writing – original draft, Validation, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Mizhi Feng:** Writing – review & editing, Writing – original draft, Resources, Formal analysis, Data curation. **Xuechen Zhang:** Writing – review & editing, Writing – original draft, Investigation, Data curation. **Jake Kaner:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This study was supported by the “Scientific Research Support” project provided by Kingfar International Inc. Thanks to the research technical team and the ErgoLAB Man-Machine Environment Testing Cloud Platform (ErgoLAB V3.0) for the related scientific research equipment support provided by the Kingfar project team. This work was also supported by the 2022 International Cooperation Joint Laboratory for Production, Education, Research, and Application of Ecological Health Care on Home Furnishing (Grant No. 20220602) and Qinglan Project of Jiangsu Province of China (Grant No. 2022QL06).

Appendix A. Appendix-summary of comparative analysis of restorative characteristics of home environment types

Based on the three design principles of material selection, color perception and personalized space configuration for healthcare housing, the design of the housing space can be divided into three types of restorative housing spaces: functionally efficient type, aesthetic display type

Table A.3
Comparative summary and analysis of restorative feature points of different home environment types.

	Capable-oriented home	Aesthetic-oriented home	Relaxed-oriented home
Style type	Modern minimalist/industrial style	Modern art/soft luxury style	Nordic/Japanese/Nature style
Core features	High space utilization rate, Emphasizes efficiency and practicality, Focuses on functionality	Visual aesthetics as the core, Artistic and personalized design, Emphasis on sensory experience	Focus on comfort, Emphasize nature and relaxation, Weaken functional zoning
Layout characteristics	Compact vertical layout, Clear flow zoning	Combination of blank space and focal area, Flowing spatial structure	Low-density layout, Soft partitions blur functional boundaries
Material selection	Glass, Resin material	Marble and stone slabs, Leather/Velvet	Logs and cotton and linen fabrics, Matte stone
Color matching	The dominant color is cool tones (gray, white, navy blue), It emphasizes the division of functional areas through contrasting color blocks.	High-saturation accent colors (Burgundy red, dark green), Transition of gold and silver metallic colors, Gradient artistic effect	Earth tones (Off-white, light brown), Low-saturation Morandi colors, Natural texture retained

Fig. A.11. Different types of home type visual stimulation materials: (a) Capable-oriented home; (b) Aesthetic-oriented home; (c) Relaxed-oriented home.

and comfortable and leisure type. The specific characteristics of each type of space are as follows.

(1) Capability-oriented home
Emphasizing efficient use of space and adaptability, this type of residence is suitable for residents with a fast-paced lifestyle. Through compact vertical layouts and multi-functional furniture (such as folding tables, hidden storage, etc.), this spatial design can maximize the utilization of limited space to meet the diverse needs of the residents. Reasonable lighting and color temperature design can also enhance the visual comfort of the space, boost work efficiency, and reduce the stress caused by spatial limitations. This type of space is commonly found in urban small-sized apartments and shared office areas, providing an efficient, flexible, and highly functional environment that helps residents better cope with the fast-paced life.

(2) Aesthetic-oriented home
Centered on visual beauty, this type of residence emphasizes artistry and sensory experience. By using high-end materials such as marble, leather, and velvet, as well as high-saturation accent colors and gradient artistic effects, a unique and visually impactful space is created. This type of residence uses an artistic spatial atmosphere to help residents regulate their emotions, reduce anxiety and stress. The combination of open floor plans and blank areas provides fluidity, enhancing the flexibility and comfort of the space. The aesthetic display residential space is suitable for use in art galleries, high-end commercial spaces, and creative studios, etc., using artistic design to enhance the emotional experience and mental health of the residents.

(3) Relaxation-oriented home
The comfortable and leisure residential type focuses on comfort and restorative qualities, aiming to provide a relaxing and restorative environment for residents. Through the introduction of soft partitions (such as carpets and green plants) and natural elements (such as wood cotton and linen fabrics), this residential space creates a warm and peaceful atmosphere, helping residents relieve stress and promote physical and mental recovery. The use of earthy color schemes and low-saturation moondust tones further enhances the sense of relaxation. The comfortable and leisure residential space is typically suitable for suburban vacation houses, healing centers, and family leisure areas, providing an ideal environment to alleviate life stress and help residents achieve emotional recovery and relaxation.

Appendix B. Appendix-multivariate analysis of variance of psychophysiological indicators

Results from the multifactor analysis of variance on physiological indicators showed that sound type produced significant differences in α, β, γ, and θ EEG bands ($p < 0.01$), with the largest effect size ($\eta^2 = 0.429$) in the γ band, and a similar tendency appeared in the subjective indicators. Home environment type also had significant effects across EEG bands ($p = 0.034$–0.045), although with smaller effect sizes ($\eta^2 = 0.052$–0.109). For the autonomic nervous system (ANS) indices, SDNN, RMSSD, PNN50, PNN20, and LF/HF were all significantly influenced by soundscape type ($p < 0.01$), and skin conductance level (SCL) was likewise significant ($p < 0.001$). Home environment type showed significant effects only for SDNN ($p = 0.027$, $\eta^2 = 0.063$) and SCL ($p < 0.001$, $\eta^2 = 0.228$). Analysis of interaction effects indicated a significant soundscape × home type interaction for β ($p < 0.001$) and γ power ($p = 0.031$), but not for α or θ. Within the autonomic measures, SCL exhibited the strongest interaction effect ($p < 0.001$), and PNN20 also showed a significant interaction ($p = 0.009$); other indicators were not significant. For variables with significant effects, LSD post hoc tests were conducted to clarify between-group differences.

The analysis of variance for the subjective indicators showed that sound type produced significant differences across all eight perceptual dimensions: comfort ($p < 0.001$), pleasant ($F = 8.891$, $p < 0.001$), perceived loudness ($p = 0.001$), preference ($F = 15.093$, $p < 0.001$), restorativeness ($p = 0.002$), roughness ($p < 0.001$), confusion ($p < 0.001$), and harmony ($p < 0.001$). These results indicate that soundscape exposure played a significant role in shaping perceived comfort, affective pleasant, perceived therapeutic benefit, and the sense of environmental coherence. The home environment type showed significant differences in seven of the eight subjective dimensions: comfort

Table B.4
Results of variance analysis of physiological indicators.

Dependent variable	Source	df	Mean square	F	P	Partial Eta Square
α	Sound	2	15.668	5.627	0.005**	0.088
	Home	2	8.866	3.184	0.045*	0.052
	Sound*Home	4	2.128	0.764	0.551	0.025
β	Sound	2	67.315	31.903	<0.001***	0.353
	Home	2	7.051	3.342	0.039*	0.054
	Sound*Home	4	11.410	5.403	<0.001***	0.156
γ	Sound	2	113.593	43.891	<0.001***	0.429
	Home	2	18.514	7.154	0.001***	0.109
	Sound*Home	4	7.143	2.76	0.031*	0.086
θ	Sound	2	84.478	31.836	<0.001***	0.352
	Home	2	9.278	3.497	0.034*	0.056
	Sound*Home	4	1.283	0.484	0.748	0.016
ΔSDNN	Sound	2	4515.339	84.667	<0.001***	0.604
	Home	2	199.987	3.75	0.027*	0.063
	Sound*Home	4	29.361	0.551	0.699	0.019
ΔRMSSD	Sound	2	237.708	6.623	0.002**	0.107
	Home	2	103.805	2.892	0.060	0.050
	Sound*Home	4	71.388	1.989	0.101	0.067
ΔPNN50	Sound	2	507.779	33.621	<0.001***	0.377
	Home	2	2.087	0.138	0.871	0.002
	Sound*Home	4	18.101	1.198	0.316	0.041
ΔPNN20	Sound	2	480.061	39.605	<0.001***	0.416
	Home	2	32.396	2.673	0.074	0.046
	Sound*Home	4	43.236	3.567	0.009**	0.114
ΔLF/HF	Sound	2	502.291	24.27	<0.001***	0.304
	Home	2	26.964	1.303	0.276	0.023
	Sound*Home	4	26.971	1.303	0.273	0.045
ΔSCL	Sound	2	30.498	42.246	<0.001***	0.432
	Home	2	11.846	16.409	<0.001***	0.228
	Sound*Home	4	23.635	32.739	<0.001***	0.541

Note: (*) $0.01 < p < 0.05$, (**) $0.001 < p < 0.01$, (***) $0.0001 < p < 0.001$, (>***) $p < 0.0001$.

Table B.5
Results of variance analysis of psychological indicators.

Inter-subject effect test Source	Dependent variable	III Class sum of squares	l Degrees of freedom	Mean square	F	Significance
Sound type	Comfort	45.19	2	22.595	16.642	<0.001
	Pleasant	29.19	2	14.595	8.891	<0.001
	Perceived loudness	11.444	2	5.722	7.486	0.001
	Preference	34.333	2	17.167	15.093	<0.001
	Restorativeness	19.825	2	9.913	6.611	0.002
	Roughness	21.016	2	10.508	11.162	<0.001
	Confusion	37.397	2	18.698	20.542	<0.001
	Harmony	80.111	2	40.056	40.476	<0.001
Home environment	Comfort	28.048	2	14.024	10.329	<0.001
	Pleasant	7.19	2	3.595	2.19	0.116
	Perceived loudness	23.444	2	11.722	15.336	<0.001
	Preference	34.429	2	17.214	15.135	<0.001
	Restorativeness	51.873	2	25.937	17.298	<0.001
	Roughness	13.349	2	6.675	7.09	0.001
	Confusion	20.206	2	10.103	11.099	<0.001
	Harmony	69.063	2	34.532	34.894	<0.001
Sound * Home	Comfort	17.619	4	4.405	3.244	0.015
	Pleasant	0.476	4	0.119	0.073	0.99
	Perceived loudness	4.222	4	1.056	1.381	0.245
	Preference	19.381	4	4.845	4.26	0.003
	Restorativeness	9.508	4	2.377	1.585	0.183
	Roughness	10.794	4	2.698	2.866	0.026
	Confusion	7.27	4	1.817	1.997	0.1
	Harmony	32.032	4	8.008	8.092	<0.001

($p < 0.001$), perceived loudness ($p < 0.001$), preference ($p < 0.001$), restorativeness ($p < 0.001$), roughness ($p = 0.001$), confusion ($p < 0.001$), and harmony ($p < 0.001$). Only pleasant did not reach significance ($p = 0.116$). In addition, significant soundscape × home environment interaction effects were observed for comfort ($p = 0.015$), preference ($p = 0.003$), roughness ($p = 0.026$), and harmony ($p < 0.001$). No significant interactions were found for pleasant, perceived loudness, restorativeness, or confusion.

Appendix C. Appendix-comprehensive analysis of the impact of soundscapes on physiological indicators

This study conducted a directional comparison of electroencephalogram (EEG) and autonomic nervous system indicators from previous work, in order to explain the significance of each indicator in recovering under different sounds and family environments. It is used to interpret the restorative meaning of each indicator under different sounds and home settings.

Overall, the present study shows a clear and consistent pattern along the sound dimension. Natural sound (N) and music sound (M) are both associated with increases in α, θ and γ power and a decrease in β power. At the same time, HRV indices such as SDNN, RMSSD, PNN50 and PNN20 increase, and the LF/HF ratio decreases. This pattern indicates stronger parasympathetic activation and more complete psycho-physiological recovery. In contrast, the urban sound (U) shows the opposite trend: β increases, θ and γ decrease, HRV indices are reduced, and LF/HF rises, suggesting sustained sympathetic activation and impaired recovery.

These directional results are highly consistent with many previous studies. For example, Choi et al. [79] reported higher α and θ and lower β in no-stress conditions, and higher β under stress in combined environmental settings. Grassini et al. [98] and Koivisto et al. [80] found increases in α (and θ in some contexts) during nature viewing or nature-related experiences. Li et al. [66], Jeon et al. [82], and Zhou et al. [86] reported enhanced HRV (e.g., higher SDNN and RMSSD) and more favourable autonomic balance (e.g., lower LF/HF) in natural or biophilic audio-visual contexts. In addition, review and methodological work by Peabody et al. [85] and Pereira et al. [83] support the view that low-stress states are typically characterized by higher RMSSD, PNN50 and PNN20 together with lower LF/HF.

Along the home-environment dimension, the present study shows that relaxed (R) and aesthetic (A) interiors perform better overall than capable interiors (C). R and A more often display a "restorative" profile, with lower β, higher γ, higher HRV and lower LF/HF. In contrast, C is more likely to show elevated β and LF/HF and reduced γ, suggesting that function-oriented spaces tend to maintain task-related arousal rather than support recovery.

It is important to note that sympathetic-related indices such as SCL show stronger context dependence. In the present study, SCL increased across different sounds and spaces, with the largest increases under music sound and in aesthetic interiors. This differs from the commonly reported decrease of SCL in relaxed states, but it is not necessarily contradictory. Higher SCL may reflect "positive arousal" rather than negative stress. Therefore, SCL should be interpreted together with EEG and HRV, rather than in isolation. This joint interpretation is consistent with the conclusions of Koivisto et al. [88] and Grassini et al. [98].

In conclusion, this comparative table not only supports the overall conclusions of the present study—namely that natural and music sounds are more restorative than urban sounds, and comfort/aesthetic spaces are more restorative than purely functional spaces—but also shows that the directional consistency of EEG and HRV provides robust cross-study evidence for recovery effects. At the same time, differences in SCL and related indices highlight that restoration does not simply mean a uniformly low-arousal state. Instead, the restorative impact of sounds and spaces on stress recovery should be understood within a multi-indicator framework.

Table C.6
Comparison of changes in physiological indicators.

		α	β	θ	γ	SCL/SC	SDNN	RMSSD	PNN50	PNN20	LF/HF
This study	N	↑	↓	↑	↑	↑	↑	↑	↑	↑	↓
	U	↑	↑	↓	↓	↑	↓	↓	↓	↓	↑
	M	↑	↓	↑	↑	↑	↑	↑	↑	↑	↓
	C	↑	↑	↑	↓	↑	↑	↑	↑	↑	↑
	A	↑	↓	↑	↑	↑	↑	↑	↑	↑	↓
	R	↑	↓	↑	↑	↑	↑	↑	↑	↑	↓
Previous researchs	Daniel K Brown et al., [81] 2013	—	—	—	—	—	—	↑(N>U)	—	—	—
	Choi et al., [79] 2015	↑(No pressure > Pressure)	↑(Pressure > No Pressure)	↑(No pressure > Pressure)	—	—	—	—	—	—	—
	Pereira [83] 2017	—	—	—	—	—	—	—	↑(Low pressure)	↑(Low pressure)	—
	Hyun-ju Jo et al., [84] 2019	—	—	—	—	—	—	—	—	—	↓(N<U)
	Li et al., [66] 2021	—	—	—	—	↓(N>U)	↑(N>U)	↑	—	—	↓
	Grassini et al., [98] 2022	↑(N>U)	—	—	—	↑(N>U)	—	—	—	—	—
	Vanhollebeke et al., [63] 2022	↑(Relax)	↑(Pressure)	↑(Context-dependent)	—	—	—	—	—	—	—
	Jeon et al., [82] 2023	—	—	—	—	—	↑(N>U)	—	—	—	—
	Peabody et al., [85] 2023	—	—	—	—	—	↑(Low pressure)	↑(Low pressure)	↑(Low pressure)	—	↓(Low pressure)
	Mika Koivisto et al., [80] 2024	↑(N)	—	↑(N)	—	—	—	—	—	—	—
	Koivisto et al. [88]	—	—	—	—	↑(N>U)	↑(N>U)	—	—	—	—
	Zhou et al., [86] 2025	—	—	—	—	↓(N)	↑(N)	↑(N)	↑(N)	—	↓(N)

↑-Indicators increase, ↓-Indicators decrease,—-Not reported, N-Nature, U-Urban, M-Music, C-Capable-oriented home, A-Aesthetic-oriented home, R-Relaxed-oriented home.

Data availability

Data will be made available upon request.

References

[1] A. Singh, M. Arora, V. Sharma, A. Kotwal, Stress: prevalence and correlates among residents of a suburban area, Ind. Psychiatr. J. 28 (2019) 98–102, https://doi.org/10.4103/ipj.ipj_33_18

[2] F. Lederbogen, P. Kirsch, L. Haddad, F. Streit, H. Tost, P. Schuch, S. Wüst, J.C. Pruessner, M. Rietschel, M. Deuschle, et al., City living and urban upbringing affect neural social stress processing in humans, Nature 474 (2011) 498–501, https://doi.org/10.1038/nature10190

[3] Y. Cai, R. Ramakrishnan, K. Rahimi, Long-term exposure to traffic noise and mortality: a systematic review and meta-analysis of epidemiological evidence between 2000 and 2020, Environ. Pollut. 269 (2021) 116222, https://doi.org/10.1016/j.envpol.2020.116222

[4] X. Zhang, S. Zhou, Building a city with low noise pollution: exploring the mental health effect thresholds of spatiotemporal environmental noise exposure and urban planning solution, Int. J. Environ. Res. Public Health 20 (2023) 4222, https://doi.org/10.3390/ijerph20054222

[5] W. Dong, D. Dai, P. Shen, R. Zhang, M. Liu, How public urban space enhance restoration benefits through combined multisensory effects: a systematic review, Land 13 (2024) 2018, https://doi.org/10.3390/land13122018

[6] R. Wang, Z. Feng, J. Pearce, Neighbourhood greenspace quantity, quality and socioeconomic inequalities in mental health, Cities 129 (2022) 103815, https://doi.org/10.1016/j.cities.2022.103815

[7] J. Kang, F. Aletta, T.T. Gjestland, L.A. Brown, D. Botteldooren, B. Schulte-Fortkamp, P. Lercher, I. van Kamp, K. Genuit, A. Fiebig, et al., Ten questions on the soundscapes of the built environment, Build. Environ. 108 (2016) 284–294, https://doi.org/10.1016/j.buildenv.2016.08.011

[8] F. Aletta, J. Kang, Ö. Axelsson, Soundscape descriptors and a conceptual framework for developing predictive soundscape models, Landsc. Urban Plan. 149 (2016) 65–74, https://doi.org/10.1016/j.landurbplan.2016.02.001

[9] M. Adams, T. Cox, G. Moore, B. Croxford, M. Refaee, S. Sharples, Sustainable soundscapes: noise policy and the urban experience, Urban Stud. 43 (2006) 2385–2398, https://doi.org/10.1080/00420980600972504

[10] B. Szeremeta, P.H.T. Zannin, Analysis and evaluation of soundscapes in public parks through interviews and measurement of noise, Sci. Total Environ. 407 (2009) 6143–6149, https://doi.org/10.1016/j.scitotenv.2009.08.039

[11] A. Brown, Soundscapes and environmental noise management, Noise Control Eng. J. 58 (2010) 493–500, https://doi.org/10.3397/1.3484178

[12] J. Kang, B. Schulte-Fortkamp, Soundscape and the built environment, vol. 525, CRC Press Boca, Raton, FL, USA, 2016.

[13] F. Aletta, T. Oberman, J. Kang, Positive health-related effects of perceiving urban soundscapes: a systematic review, Lancet 392 (2018) S3, https://doi.org/10.1016/S0140-6736(18)32044-0

[14] E. Abdelmoula, B. Abdelmoula, N.B. Abdelmoula, Natural soundscapes, urban design and psychological well-being, Eur. Psychiatry 67 (2024) S680–S680, https://doi.org/10.1192/j.eurpsy.2024.1414

[15] C.-H. Hsieh, J.-Y. Yang, C.-W. Huang, W.C.B. Chin, The effect of water sound level in virtual reality: a study of restorative benefits in young adults through immersive natural environments, J. Environ. Psychol. 88 (2023) 102012, https://doi.org/10.1016/j.jenvp.2023.102012

[16] B. Lam, Z.-T. Ong, K. Ooi, W.-H. Ong, T. Wong, K.N. Watcharasupat, V. Boey, I. Lee, J.Y. Hong, J. Kang, et al., Automating urban soundscape enhancements with AI: in-situ assessment of quality and restorativeness in traffic-exposed residential areas, Build. Environ. 266 (2024) 112106, https://doi.org/10.1016/j.buildenv.2024.112106

[17] J.Y. Hong, Z.-T. Ong, B. Lam, K. Ooi, W.-S. Gan, J. Kang, J. Feng, S.-T. Tan, Effects of adding natural sounds to urban noises on the perceived loudness of noise and soundscape quality, Sci. Total Environ. 711 (2020) 134571, https://doi.org/10.1016/j.scitotenv.2019.134571

[18] M. Saskovets, I. Saponkova, Z. Liang, Effects of sound interventions on the mental stress response in adults: scoping review, JMIR Ment. Health 12 (2025) e69120, https://preprints.jmir.org/preprint/69120.

[19] S. Paszkiel, P. Dobrakowski, A. Łysiak, The impact of different sounds on stress level in the context of EEG, cardiac measures and subjective stress level: a pilot study, Brain Sci. 10 (2020) 728, https://doi.org/10.3390/brainsci10100728

[20] Z. Li, J. Kang, Sensitivity analysis of changes in human physiological indicators observed in soundscapes, Landsc. Urban Plan. 190 (2019) 103593, https://doi.org/10.1016/j.landurbplan.2019.103593

[21] G.M.E. Sanchez, T. Van Renterghem, K. Sun, B. De Coensel, D. Botteldooren, Using virtual reality for assessing the role of noise in the audio-visual design of an urban public space, Landsc. Urban Plan. 167 (2017) 98–107, https://doi.org/10.1016/j.landurbplan.2017.05.018

[22] J.Y. Jeon, H.I. Jo, B.B. Santika, H. Lee, Crossed effects of audio-visual environment on indoor soundscape perception for pleasant open-plan office environments, Build. Environ. 207 (2022) 108512, https://doi.org/10.1016/j.buildenv.2021.108512

[23] H. Touhami, D. Berkouk, T.A.K. Bouzir, S. Khelil, M.M. Gomaa, The influence of multisensory perception on student outdoor comfort in university campus design, Atmosphere 16 (2025) 150, https://doi.org/10.3390/atmos16020150

[24] T. Van Renterghem, A. Lippens, The audio-visual incongruency asymmetry. natural sounds in an urban visual setting are more relaxing than urban sounds in visual nature, Urban For. Urban Green. 101 (2024) 128514, https://doi.org/10.1016/j.ufug.2024.128514

[25] R.S. Ulrich, Effects of interior design on wellness: theory and recent scientific research, in: Journal of Health Care Interior Design Proceedings from the.symposium on Health Care Interior Design.symposium on Health Care Interior Design, vol. 3, 1991, pp. 97.

[26] S. Kaplan, The restorative benefits of nature: toward an integrative framework, J. Environ. Psychol. 15 (1995) 169–182, https://doi.org/10.1016/0272-4944(95)90001-2

[27] A.K. Demirkol, A.K. Onac, Integrating biophilic design elements into office designs, AIN Shams Eng. J. 15 (2024), https://doi.org/10.1016/j.asej.2024.102962

[28] J. Zhao, Y. Nagai, W. Gao, T. Shen, Y. Fan, The effects of interior materials on the restorativeness of home environments, Int. J. Environ. Res. Public Health 20 (2023) 6364, https://doi.org/10.3390/ijerph20146364

[29] J. Yin, J. Yuan, N. Arfaei, P.J. Catalano, J.G. Allen, J.D. Spengler, Effects of biophilic indoor environment on stress and anxiety recovery: a between-subjects experiment in virtual reality, Environ. Int. 136 (2020) 105427, https://doi.org/10.1016/j.envint.2019.105427

[30] D. Kopec, Environmental psychology for design, Bloomsbury Publishing USA, 2018.

[31] J. Wang, Y. Fang, Y. Huang, J. Chen, X. Luo, Study on the effect of home night lighting environment on students' eyestrain and visual efficacy, in: E3S Web of Conferences, vol. 261, EDP Sciences, 2021, https://doi.org/10.1051/e3sconf/202126103007, p. 03007.

[32] A.S.L. Yong, D. Haines, L.H. Joseph, Home environment design theories and models related to the occupational performance, participation and well-being of people with intellectual disabilities: a scoping review, Br. J. Occup. Ther. 86 (2023) 665–677, https://doi.org/10.1177/03080226231183291

[33] B. Boegheim, R. Appel-Meulenbroek, D. Yang, M. Loomans, Indoor environmental quality (IEQ) in the home workplace in relation to mental well-being, Facilities 40 (2022) 125–140, https://doi.org/10.1108/F-05-2022-0070

[34] H.I. Jo, J.Y. Jeon, Influence of indoor soundscape perception based on audiovisual contents on work-related quality with preference and perceived productivity in open-plan offices, Build. Environ. 208 (2022) 108598, https://doi.org/10.1016/j.buildenv.2021.108598

[35] Q. Meng, J. Kang, H. Jin, Field study on the influence of spatial and environmental characteristics on the evaluation of subjective loudness and acoustic comfort in underground shopping streets, Appl. Acoust. 74 (2013) 1001–1009, https://doi.org/10.1016/j.apacoust.2013.02.003

[36] S. Viollon, C. Lavandier, C. Drake, Influence of visual setting on sound ratings in an urban environment, Appl. Acoust. 63 (2002) 493–511, https://doi.org/10.1016/S0003-682X(01)00053-6

[37] S. Torresin, E. Ratcliffe, F. Aletta, R. Albatici, F. Babich, T. Oberman, J. Kang, The actual and ideal indoor soundscape for work, relaxation, physical and sexual activity at home: a case study during the COVID-19 lockdown in London, Front. Psychol. 13 (2022) 1038303, https://doi.org/10.3389/fpsyg.2022.1038303

[38] G. Liu, J. Zou, M. Qiao, H. Zhu, Y. Yang, H. Guan, S. Hu, Stress recovery at home: effects of the indoor visual and auditory stimuli in buildings, Build. Environ. 244 (2023) 110752, https://doi.org/10.1016/j.buildenv.2023.110752

[39] B. Sona, E. Dietl, A. Steidle, Recovery in sensory-enriched break environments: integrating vision, sound and scent into simulated indoor and outdoor environments, Ergonomics 62 (2019) 521–536, https://doi.org/10.1080/00140139.2018.1491643

[40] Y. Lu, J.K.A. Tan, Y. Hasegawa, S.-K. Lau, The interactive effects of traffic sound and window views on indoor soundscape perceptions in the residential area, J. Acoust. Soc. Am. 153 (2023) 972–989, https://doi.org/10.1121/10.0017203

[41] M. Erfanian, A. Mitchell, F. Aletta, J. Kang, Psychological well-being and demographic factors can mediate soundscape pleasantness and eventfulness: a large sample study, J. Environ. Psychol. 77 (2021) 101660, https://doi.org/10.1016/j.jenvp.2021.101660

[42] R. Wang, J. Zhao, A good sound in the right place: exploring the effects of auditory-visual combinations on aesthetic preference, Urban For. Urban Green. 43 (2019) 126356, https://doi.org/10.1016/j.ufug.2019.05.018

[43] G. Aniansson, K. Pettersson, Y. Peterson, Traffic noise annoyance and noise sensitivity in persons with normal and impaired hearing, J. Sound Vib. 88 (1983) 85–97, https://doi.org/10.1016/0022-460X(83)90680-6

[44] Z. Chen, C. von Haaren, Urban dwellers' perceptions of soundscapes and landscapes: Impacts of birdsongs and noises, and moderation by demographic characteristics, Urban Forestry & Urban Greening, 2025, https://doi.org/10.1016/j.ufug.2025.128838 128838.

[45] S. Shimai, K. Fukuda, M. Terasaki, Pleasantness-unpleasantness of environmental sounds and gender difference in evaluation, Percept. Mot. Skills 76 (1993) 635–640, https://doi.org/10.2466/pms.1993.76.2.635

[46] A.M. Abbasi, E. Darvishi, M.A. Rodrigues, K. Sayehmiri, Gender differences in cognitive performance and psychophysiological responses during noise exposure and different workloads, Appl. Acoust. 189 (2022) 108602, https://doi.org/10.1016/j.apacoust.2021.108602

[47] J.Y. Jeon, H.I. Jo, Effects of audio-visual interactions on soundscape and landscape perception and their influence on satisfaction with the urban environment, Build. Environ. 169 (2020) 106544, https://doi.org/10.1016/j.buildenv.2019.106544

[48] F.M. Wartmann, J. Frick, F. Kienast, M. Hunziker, Factors influencing visual landscape quality perceived by the public. results from a national survey, Landsc. Urban Plan. 208 (2021) 104024, https://doi.org/10.1016/j.landurbplan.2020.104024

[49] C. Tarlao, J. Steffens, C. Guastavino, Investigating contextual influences on urban soundscape evaluations with structural equation modeling, Build. Environ. 188 (2021) 107490, https://doi.org/10.1016/j.buildenv.2020.107490

[50] J. Cohen, A power primer, Am. Psychol. Assoc. 4 (2016) 279–284. https://doi.org/10.1037/14805-018.,

[51] J.Y. Hong, J.Y. Jeon, The effects of audio–visual factors on perceptions of environmental noise barrier performance, Landsc. Urban Plan. 125 (2014) 28–37, https://doi.org/10.1016/j.landurbplan.2014.02.001

[52] X. Fei, Y. Wu, J. Dong, D. Kong, The effects of soundscape interactions on the restorative potential of urban green spaces, Sustainability 17 (2025) 2674, https://doi.org/10.3390/su17062674

[53] C. Liu, X. Jing, J. Shi, J. Li, Y. Zhang, W. Gao, Effects of natural sound on human stress recovery based on EEG techniques, J. Environ. Psychol. 97 (2024) 102365, https://doi.org/10.1016/j.jenvp.2024.102365

[54] K.W. Ma, C.M. Mak, H.M. Wong, Effects of environmental sound quality on soundscape preference in a public urban space, Appl. Acoust. 171 (2021) 107570, https://doi.org/10.1016/j.jenvp.2024.102365

[55] E. Labbé, N. Schmidt, J. Babin, M. Pharr, Coping with stress: the effectiveness of different types of music, Appl. Psychophysiol. Biofeedback 32 (2007) 163–168, https://doi.org/10.1007/s10484-007-9043-9

[56] P.-R. Kong, K.-T. Han, Psychological and physiological effects of soundscapes: a systematic review of 25 experiments in the english and Chinese literature, Sci. Total Environ. 929 (2024) 172197, https://doi.org/10.1016/j.scitotenv.2024.172197

[57] G. Pei, T. Li, A literature review of EEG-based affective computing in marketing, Front. Psychol. 12 (2021) 602843, https://doi.org/10.3389/fpsyg.2021.602843

[58] X. Li, Y. Zhang, P. Tiwari, D. Song, B. Hu, M. Yang, Z. Zhao, N. Kumar, P. Marttinen, EEG based emotion recognition: a tutorial and review, ACM Comput. Surv. 55 (2022) 1–57, https://doi.org/10.1145/3524499

[59] Z. Yang, Y. Zhang, J. Ning, X. Wang, Z. Wu, Early diagnosis of autism: a review of video-based motion analysis and deep learning techniques, IEEE Access (2025), https://doi.org/10.1109/ACCESS.2024.3523872

[60] J. Körmendi, E. Ferentzi, B. Weiss, Z. Nagy, Topography of movement-related delta and theta brain oscillations, Brain Topogr. 34 (2021) 608–617, https://doi.org/10.1007/s10548-021-00854-0

[61] Y.-C. Chiang, D. Li, H.-A. Jane, Wild or tended nature? the effects of landscape location and vegetation density on physiological and psychological responses, Landsc. Urban Plan. 167 (2017) 72–83, https://doi.org/10.1016/j.landurbplan.2017.06.001

[62] M. Gola, M. Magnuski, I. Szumska, A. Wróbel, EEG beta band activity is related to attention and attentional deficits in the visual performance of elderly subjects, Int. J. Psychophysiol. 89 (2013) 334–341, https://doi.org/10.1016/j.ijpsycho.2013.05.007

[63] G. Vanhollebeke, S. De Smet, R. De Raedt, C. Baeken, P. van Mierlo, M.-A. Vanderhasselt, The neural correlates of psychosocial stress: a systematic review and meta-analysis of spectral analysis EEG studies, Neurobiol. Stress 18 (2022) 100452, https://doi.org/10.1016/j.ynstr.2022.100452

[64] T. Laohakangvalvit, P. Sripian, Y. Nakagawa, C. Feng, T. Tazawa, S. Sakai, M. Sugaya, Study on the psychological states of olfactory stimuli using electroencephalography and heart rate variability, Sensors 23 (2023) 4026, https://doi.org/10.3390/s23084026

[65] N. Gullett, Z. Zajkowska, A. Walsh, R. Harper, V. Mondelli, Heart rate variability (HRV) as a way to understand associations between the autonomic nervous system (ANS) and affective states: a critical review of the literature, Int. J. Psychophysiol. 192 (2023) 35–42, https://doi.org/10.1016/j.ijpsycho.2023.08.001

[66] Z. Li, M. Ba, J. Kang, Physiological indicators and subjective restorativeness with audio-visual interactions in urban soundscapes, Sustain. Cities Soc. 75 (2021) 103360, https://doi.org/10.1016/j.scs.2021.103360

[67] A. Burlacu, C. Brinza, A. Brezulianu, A. Covic, Accurate and early detection of sleepiness, fatigue and stress levels in drivers through heart rate variability parameters: a systematic review, Rev. Cardiovasc. Med. 22 (2021) 845–852, https://doi.org/10.31083/j.rcm2203090

[68] A.G. Srinivasan, S.S. Smith, C.L. Pattinson, D. Mann, K. Sullivan, P. Salmon, S.S. Soleimanloo, Heart rate variability as an indicator of fatigue: a structural equation model approach, Transp. Res. Part F Traffic Psychol. Behav. 103 (2024) 420–429, https://doi.org/10.1016/j.trf.2024.04.015

[69] B. Yang, H. Yao, P. Yang, Y. Guo, F. Wang, C. Yang, A. Li, L. Che, Effects of thermal and acoustic environments on workers' psychological and physiological stress in deep underground spaces, Build. Environ. 212 (2022) 108830, https://doi.org/10.1016/j.buildenv.2022.108830

[70] R. Li, R. Yan, W. Cheng, H. Ren, Effect of resistance training on heart rate variability of anxious female college students, Front. Public Health 10 (2022) 1050469, https://doi.org/10.3389/fpubh.2022.1050469

[71] S. Solhjoo, M.C. Haigney, E. McBee, J.J. van Merrienboer, L. Schuwirth, A.R. Artino Jr, A. Battista, T.A. Ratcliffe, H.D. Lee, S.J. Durning, Heart rate and heart rate variability correlate with clinical reasoning performance and self-reported measures of cognitive load, Sci. Rep. 9 (2019) 14668, https://doi.org/10.1038/s41598-019-50280-3

[72] R.L. Mandryk, K.M. Inkpen, T.W. Calvert, Using psychophysiological techniques to measure user experience with entertainment technologies, Behav. Inf. Technol. 25 (2006) 141–158, https://doi.org/10.1080/01449290500331156

[73] E. Fasllija, L. Shtrepi, E. Alimadhi, Development of the Albanian translation of the perceived affective qualities in the ISO/TS 12913-2 soundscape standard, Appl. Acoust. 217 (2024) 109835, https://doi.org/10.1016/j.apacoust.2023.109835

[74] J. Kang, Q. Meng, H. Jin, Effects of individual sound sources on the subjective loudness and acoustic comfort in underground shopping streets, Sci. Total Environ. 435 (2012) 80–89, https://doi.org/10.1016/j.scitotenv.2012.06.105

[75] T. Hartig, Validation of a measure of perceived environmental restorativeness, Goteborg Psychol. Rep. 26 (1996).

[76] M. Erfanian, A.J. Mitchell, J. Kang, F. Aletta, The psychophysiological implications of soundscape: a systematic review of empirical literature and a research agenda, Int. J. Environ. Res. Public Health 16 (2019) 3533, https://doi.org/10.3390/ijerph16193533

[77] S.R. Payne, The production of a perceived restorativeness soundscape scale, Appl. Acoust. 74 (2013) 255–263, https://doi.org/10.1016/j.apacoust.2011.11.005

[78] S.R. Payne, C. Guastavino, Exploring the validity of the perceived restorativeness soundscape scale: a psycholinguistic approach, Front. Psychol. 9 (2018) 2224, https://doi.org/10.3389/fpsyg.2018.02224

[79] Y. Choi, M. Kim, C. Chun, Measurement of occupants' stress based on electroencephalograms (EEG) in twelve combined environments, Build. Environ. 88 (2015) 65–72, https://doi.org/10.1016/j.buildenv.2014.10.003

[80] M. Koivisto, J. Koskinen, S. Jokiaho, T. Vahanne, M. Pohjola, E. Kontio, A short simulated nature experience as an effective way to promote restoration from work-related stress, Scand. J. Psychol. 65 (2024) 954–965, https://doi.org/10.1111/sjop.13044

[81] D.K. Brown, J.L. Barton, V.F. Gladwell, Viewing nature scenes positively affects recovery of autonomic function following acute-mental stress, Environ. Sci. Technol. 47 (2013) 5562–5569, https://doi.org/10.1021/es305019p

[82] J.Y. Jeon, H.I. Jo, K. Lee, Psycho-physiological restoration with audio-visual interactions through virtual reality simulations of soundscape and landscape experiences in urban, waterfront, and green environments, Sustain. Cities Soc. 99 (2023) 104929, https://doi.org/10.1016/j.scs.2023.104929

[83] T. Pereira, P.R. Almeida, J.P. Cunha, A. Aguiar, Heart rate variability metrics for fine-grained stress level assessment, Comput. Methods Programs Biomed. 148 (2017) 71–80, https://doi.org/10.1016/j.cmpb.2017.06.018

[84] H. Jo, C. Song, H. Ikei, S. Enomoto, H. Kobayashi, Y. Miyazaki, Physiological and psychological effects of forest and urban sounds using high-resolution sound sources, Int. J. Environ. Res. Public Health 16 (2019) 2649, https://doi.org/10.3390/ijerph16152649

[85] J.E. Peabody, R. Ryznar, M.T. Ziesmann, L. Gillman, R.J. Ryznar, L.M. Gillman, A systematic review of heart rate variability as a measure of stress in medical professionals, Cureus 15 (2023), https://doi.org/10.7759/cureus.34345

[86] C. Zhou, B. Xu, X. Xu, J. Kaner, Exploring the creation of multi-modal soundscapes in the indoor environment: a study of stimulus modality and scene type affecting physiological recovery, J. Build. Eng. (2025) 113327, https://doi.org/10.1016/j.jobe.2025.113327

[87] M. Koivisto, S. Grassini, Mental imagery of nature induces positive psychological effects, Curr. Psychol. 42 (2023) 30348–30363, https://doi.org/10.1007/s12144-022-04088-6

[88] M. Koivisto, S. Grassini, The restorative effects of mental imagery of nature: a study on subjective and physiological responses, J. Environ. Psychol. 96 (2024) 102346, https://doi.org/10.1016/j.jenvp.2024.102346

[89] X. Zhang, Z. Lian, Y. Wu, Human physiological responses to wooden indoor environment, Physiol. Behav. 174 (2017) 27–34, https://doi.org/10.1016/j.physbeh.2017.02.043

[90] N. Zhang, L. Zhao, J. Shi, W. Gao, Impact of visual and textural characteristics of street walls on stress recovery, Sci. Rep. 14 (2024) 15115, https://doi.org/10.1038/s41598-024-64618-z

[91] I. Bower, R. Tucker, P.G. Enticott, Impact of built environment design on emotion measured via neurophysiological correlates and subjective indicators: a systematic review, J. Environ. Psychol. 66 (2019) 101344, https://doi.org/10.1016/j.jenvp.2019.101344

[92] S. Huang, J. Qi, W. Li, J. Dong, C.K. van den Bosch, The contribution to stress recovery and attention restoration potential of exposure to urban green spaces in low-density residential areas, Int. J. Environ. Res. Public Health 18 (2021) 8713, https://doi.org/10.3390/ijerph18168713

[93] Q. Huang, M. Yang, H.-A. Jane, S. Li, N. Bauer, Trees, grass, or concrete? the effects of different types of environments on stress reduction, Landsc. Urban Plan. 193 (2020) 103654, https://doi.org/10.1016/j.landurbplan.2019.103654

[94] Y. Zhang, L. Zhang, H. Hua, J. Jin, L. Zhu, L. Shu, X. Xu, F. Kuang, Y. Liu, Relaxation degree analysis using frontal electroencephalogram under virtual reality relaxation scenes, Front. Neurosci. 15 (2021) 719869, https://doi.org/10.3389/fnins.2021.719869

[95] C. Young, M. Hofmann, D. Frey, M. Moretti, N. Bauer, Psychological restoration in urban gardens related to garden type, biodiversity and garden-related stress, Landsc. Urban Plan. 198 (2020) 103777, https://doi.org/10.1016/j.landurbplan.2020.103777

[96] L. Zhang, P.Y. Tan, D. Richards, Relative importance of quantitative and qualitative aspects of urban green spaces in promoting health, Landsc. Urban Plan. 213 (2021) 104131, https://doi.org/10.1016/j.landurbplan.2021.104131

[97] P. Lindemann-Matthies, X. Junge, D. Matthies, The influence of plant diversity on people's perception and aesthetic appreciation of grassland vegetation, Biol. Conserv. 143 (2010) 195–202, https://doi.org/10.1016/j.biocon.2009.10.003

[98] S. Grassini, G.V. Segurini, M. Koivisto, Watching nature videos promotes physiological restoration: evidence from the modulation of alpha waves in electroencephalography, Front. Psychol. 13 (2022) 871143, https://doi.org/10.3389/fpsyg.2022.871143

[99] Y. Saito, M. Nishimaki, Y. Tatsumi, S. Ono, M. Kinoshita, S. Sasayama, K. Saito, Effects of wooden and vinyl interior finishes on stress reduction estimate by biological and psychological parameters, Mokuzai Gakkaishi/J. Jpn. Wood Res. Soc. 55 (2009) 101–107, https://doi.org/10.2488/jwrs.55.101

[100] C. Valentine, The impact of architectural form on physiological stress: a systematic review, Front. Comput. Sci. 5 (2024) 1237531, https://doi.org/10.3389/fcomp.2023.1237531

[101] S. Korpilo, E. Nyberg, K. Vierikko, A. Ojala, J. Kaseva, J. Lehtimäki, L. Kopperoinen, G. Cerwén, M. Hedblom, E. Castellazzi, et al., Landscape and soundscape quality promote stress recovery in nearby urban nature: a multisensory field experiment, Urban For. Urban Green. 95 (2024) 128286, https://doi.org/10.1016/j.ufug.2024.128286

[102] Z. Li, K. Wang, M. Hai, P. Cai, Y. Zhang, Preliminary study on gender differences in EEG-based emotional responses in virtual architectural environments, Buildings 14 (2024) 2884, https://doi.org/10.3390/buildings14092884

[103] L. Kazen, G. Akbulut, B. Wagaman, E. Choudhury, S. Pass, E. Zamzow, The effects of music genre on physiological measures of stress from arithmetic testing, 2017, https://doi.org/10.1093/jmt/39.2.101

[104] J.J. Alvarsson, S. Wiens, M.E. Nilsson, Stress recovery during exposure to nature sound and environmental noise, Int. J. Environ. Res. Public Health 7 (2010) 1036–1046, https://doi.org/10.3390/ijerph7031036

[105] A. Calderón-García, E. Álvarez-Gallardo, P. Belinchón-DeMiguel, V.J. Clemente-Suárez, Gender differences in autonomic and psychological stress responses among educators: a heart rate variability and psychological assessment study, Front. Psychol. 15 (2024) 1422709, https://doi.org/10.3389/fpsyg.2024.1422709

[106] D. Kuang, L. Cui, S. Kuang, R. Yang, X. Chen, L. Zhang, R. Lv, S. Ou, C. Song, Effect of gender-related depression on heart rate variability during an autonomic nervous test, Psychiatry Res. 272 (2019) 258–264, https://doi.org/10.1016/j.psychres.2018.12.099

[107] H.A. Abhishekh, P. Nisarga, R. Kisan, A. Meghana, S. Chandran, T. Raju, T.N. Sathyaprabha, Influence of age and gender on autonomic regulation of heart, J. Clin. Monit. Comput. 27 (2013) 259–264, https://doi.org/10.1007/s10877-012-9424-3

[108] J.C. Benítez-Agudelo, D. Restrepo, V.J. Clemente-Suárez, Gender differences in psychophysiological responses to stress and academic performance: the role of sleep, anxiety, and heart rate variability, Physiol. Behav. 299 (2025) 114970, https://doi.org/10.1016/j.physbeh.2025.114970